



**Verified Carbon
Standard**

TRAMONTANA ICS-01 – THE DISTRIBUTION OF IMPROVED COOKSTOVES TO INDIGENOUS COMMUNITIES ACROSS RURAL INDIA



Carbon
— C H E C K —

Report ID	CC IPL2454/VCS/VER/TDICSICRI/20240828
Project title	Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India
Project ID	4457
Verification period	24-August-2023 to 12-September-2024
Original date of issue	21-October-2025
Most recent date of issue	06 – January - 2026
Version	1.2
VCS Standard Version	VCS Standard v4.7

Client	Tramontana Impact Limited
Prepared by	Carbon Check (India) Private Ltd.
Approved by	Amit Anand, CEO
Work carried out by	Manas Hadler – Team Leader/ Technical Expert Temesgen Areaya – Trainee Assessor Nisha Jha – Trainee Assessor S. Ranganathan – Technical Reviewer

Summary:

- **A description of the verification of the project**

Verification:

Carbon Check (India) Private Limited, hereinafter referred to as “Carbon Check”, an accredited Validation and Verification Body (VVB), was engaged by the Project Proponent, Tramontana Impact Limited, to perform the verification of the VCS Project Activity titled “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” (VCS ID – 4457). The verification has been conducted in accordance with the requirements of the Verified Carbon Standard (VCS) Version 4.7 /B01/ and the provisions of ISO 14064-3:2019. The verification was based on a desk review of the Monitoring Report /01/, the registered/revised Project Description (PD) /03/, the emission reduction calculation spreadsheet /02/, and other supporting documentation provided by the Project Proponent. The review was supplemented by a site visit and interviews with project stakeholders to validate the implementation and monitoring systems in place. The monitoring period covered under this verification spans from 24-August-2023 to 12-September-2024, inclusive of both dates.

Project:

The project applies the approved VCS methodology VM0050, Version 1.0 /B02/, and involves the dissemination of improved cookstoves (ICS) to indigenous households across various rural regions in India. The objective of the project is to reduce greenhouse gas emissions by replacing traditional biomass-based cooking methods with energy-efficient stove technologies. The intervention contributes to lower consumption of non-renewable biomass and improved thermal efficiency, thereby achieving measurable emission reductions over the crediting period.

- **The purpose and scope of verification**

Purpose: The verification aimed to evaluate the implementation of the monitoring plan and to assess whether the monitoring methodology has been applied in accordance with the approved project documentation and the requirements of the Verified Carbon Standard (VCS) Version 4.7 /B01/. The verification further aims to ensure that the data underlying the calculation of greenhouse gas (GHG) emission reductions is complete, accurate, consistent, and transparently presented. Compliance with the registered/revised PD /03/, applied methodology (VM0050, Version 1.0) /B02/, monitoring plan, and relevant host country and VCS requirements was assessed to confirm that the project has been implemented in line with the approved design and conservative assumptions.

Scope:

The scope of this verification includes:

- Verification of project implementation and operation against the registered/revised VCS PD.
- Assessment of the conformity of the implemented monitoring plan with the registered documentation and the applied baseline and monitoring methodology.
- Verification that the actual monitoring systems, data collection procedures, and quality assurance measures are in alignment with those described in the monitoring plan.

- Evaluation of reported GHG emission reduction data and the provision of a verification conclusion with a reasonable level of assurance on the absence of material misstatements.
- Review of supporting documentation to confirm that reported data are substantiated by verifiable and reliable evidence.
- The verification exercise is intended to confirm that the reported emission reductions are conservative, complete, and eligible for issuance under the VCS framework.
- **The method and criteria used for verification**
 - (a) The desk-based assessment involved the following key activities:
 1. A detailed review of the monitoring report and supporting documentation to verify completeness and consistency of the data submitted.
 2. Assessment of the implemented monitoring plan and applied methodology, with specific attention to measurement frequencies, metering equipment specifications, calibration records, and adherence to defined QA/QC procedures.
 3. Evaluation of the data management framework and QA/QC systems with a focus on their role in ensuring the accuracy and traceability of emission reduction reporting.
 - (b) On-site Assessment, including:
 1. Evaluation of the project's physical implementation and operational status against the revised PD /03/.
 2. Review of data flow processes related to the collection, aggregation, and reporting of monitoring parameters.
 3. Interviews with project and monitoring personnel to confirm that monitoring procedures were executed in accordance with the approved monitoring plan.
 4. Cross-checks of monitoring report data against other project records such as stove distribution inventories, procurement documentation, and field survey results.
 5. Inspection of monitoring equipment and review of calibration logs to confirm compliance with methodological and monitoring requirements.
 6. Verification of emission reduction calculations, including underlying assumptions and formulae used in the calculation spreadsheet.
 7. Assessment of quality assurance and quality control procedures in place to detect, prevent, and rectify any material discrepancies in the monitoring data.
- **The number of findings raised during verification**

A risk-based approach has been followed to perform this verification. During the course of verification, a total of 16 findings were raised, which include:

Type	Count	Summary of Findings
Clarification Requests (CLs)	05	Refer to Appendix 5 for further details.

Corrective Action Requests (CARs)	09	Refer to Appendix 5 for further details.
Forward Action Requests (FARs)	02	Refer to Appendix 5 for further details.

All CARs and CLs will be resolved by the Project Proponent. Any remaining FARs from validation and/or previous verification will also be resolved during this verification.

Any uncertainties associated with the verification

The monitoring report /01/, emissions reduction calculations /02/, and supporting documents provided comply with the requirements of VCS Standard version 4.7 /B01/. The verification team identified no additional uncertainties or quality restrictions.

- **Summary of the verification conclusion**

In the opinion of Carbon Check, the emission reductions reported for the “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” project in the monitoring report version 1.3 dated 02-January-2026 are fairly and accurately stated. Accordingly, Carbon Check (India) Private Ltd. certifies that the emission reductions from the project for the period 24-August-2023 to 12-September-2024 are 172,045 tCO₂e.

CONTENTS

1	INTRODUCTION	8
1.1	Objective	8
1.2	Scope and Criteria	8
1.3	Level of Assurance.....	10
1.4	Summary Description of the Project	10
2	VERIFICATION PROCESS	11
2.1	Method and Criteria.....	11
2.2	Document Review	13
2.3	Interviews	13
2.4	Site Visits	22
2.5	Resolution of Findings	27
2.6	Eligibility for Validation Activities	28
3	VALIDATION FINDINGS	28
3.1	Methodology Deviations.....	28
3.2	Project Description Deviations.....	28
3.3	New Project Activity Instances in Grouped Projects.....	28
3.4	Baseline Reassessment	31
4	VERIFICATION FINDINGS.....	33
4.1	Project Details	33
4.2	Safeguards and Stakeholder Engagement	36
4.3	Accuracy of Reduction and Removal Calculations.....	46
4.4	Quality of Evidence to Determine Reductions and Removals.....	76
4.5	Non-Permanence Risk Analysis.....	76
5	VERIFICATION OPINION.....	77
5.1	Verification Summary	77
5.2	Verification Conclusion	78
5.3	Ex-ante vs Ex-post ERR Comparison	79
	APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION	80
	APPENDIX 2: SUPPORTING DOCUMENTS	81

APPENDIX 2.1: REFERENCE DOCUMENTS.....	81
APPENDIX 2.2: BACKGROUND REFERENCE DOCUMENTS	83
APPENDIX 3: ABBREVIATIONS.....	85
APPENDIX 4: CERTIFICATES OF COMPETENCE	86
APPENDIX 5: FINDINGS LOG	90

1 INTRODUCTION

1.1 Objective

Carbon Check has been contracted by Tramontana Impact Limited, the Project Proponent (PP), to undertake the verification of the project titled “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” for the monitoring period 24-August-2023 to 12-September-2024 (inclusive of both dates).

The primary objective of the verification is to confirm that:

- The project has been implemented as described in the revised VCS PD /03/.
- The monitoring system is operational, functioning as intended, and capable of producing verifiable GHG emission reductions.
- The data reported by the Project Proponent are accurate, complete, consistent, transparent, and free from material misstatement or omission, based on a thorough review of monitoring records, calculation spreadsheets, and supporting documentation.

The verification was conducted in accordance with the requirements of the VCS Standard Version 4.7 /B01/ and other applicable components of the VCS Program to ensure the consistency, integrity, and reliability of the verification process and its results.

1.2 Scope and Criteria

The verification of this project is based on the Monitoring Report of this monitoring period /01/, registered/revised VCS PD /03/, Emission reduction calculation spreadsheets /02/, supporting documents made available to the verifier and information collected through performing on-site interviews. Furthermore, publicly available information was considered as far as available and required.

The verification was conducted by Carbon Check (India) Private Ltd. and is based on the following sources of information:

- The registered/revised VCS PD /03/;
- The monitoring report version 1.3 dated 02-January-2026 /01/;
- Ex-post emission reduction calculation spreadsheets /02/;
- Supporting documentation submitted by the Project Proponent.
- Data and clarifications collected through on-site assessments and interviews with project stakeholders.

- Publicly available information where applicable and relevant.

A risk-based approach was employed, emphasizing areas of significant risk or uncertainty that may influence the accuracy, completeness, or conservativeness of the reported GHG emission reductions. Particular attention was given to the robustness of the monitoring system, adherence to the registered/revised PD, and conformance with applicable methodological and regulatory requirements.

The verification has been carried out against the following key criteria:

- VCS Standard Version 4.7 /B01/
- VCS Program Guide Version 4.4 /B01/
- VCS Validation and Verification Manual Version 3.2 /B01/
- Approved VCS Methodology VM0050, Version 1.0 /B02/ and relevant associated tools
- Applicable VCS Tools and Modules
- Host country legal and regulatory requirements, where relevant

The scope of this verification, by independent checking of objective evidence, is as follows:

- To confirm that the project has been implemented in accordance with the registered/revised VCS PD /03/;
- To verify compliance with the approved methodology and the applied monitoring plan;
- To assess whether the monitoring system is properly implemented and functioning to generate verifiable and conservative emission reductions;
- To ensure the reported GHG emission reductions are accurate, transparent, complete, consistent, and free of material error or omission;
- To verify that all monitored data are supported by objective and traceable evidence;
- To express a verification conclusion with a reasonable level of assurance regarding the reliability of the reported emission reductions.

The verification process comprised the following phases:

1. Completeness check and desk review of submitted documentation.
2. On-site assessment, including interviews with relevant stakeholders and field inspection.
3. Resolution of findings, if any, followed by the issuance of the Final Verification Report and VCS Verification Deed of Representation.

All verification activities were conducted in accordance with Carbon Check quality management procedures and impartiality policies. The verification team has maintained independence throughout the engagement and has not provided any form of consulting or advisory services to the Project Proponent.

1.3 Level of Assurance

The verification has been conducted with a reasonable level of assurance, in accordance with ISO 14064-3:2019 and VCS Standard v4.7 /B01/.

☒ Reasonable level of assurance

☐ Limited level of assurance

The materiality threshold for quantitative errors, omissions, or misstatements was set at one percent (1%) of the total reported GHG emission reductions, corresponding to 172,045 tCO₂e. Which is consistent with the guidance provided under section 4.1.10 of the VCS Standard version 4.7 /B01/.

Based on the verification activities, the verification body concludes that the reported emission reductions for the monitoring period are free from material error and in compliance with applicable VCS requirements.

Based on the evidence reviewed and the verification activities conducted, the Verification and Validation Body (VVB) concludes that the reported GHG emission reductions for the monitoring period are free from material misstatements and in compliance with applicable VCS requirements. The verification has been conducted independently, objectively, and in adherence to Carbon Check impartiality and quality management systems.

1.4 Summary Description of the Project

The project activity titled “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” involves a clean cooking intervention that reduces anthropogenic greenhouse gas (GHG) emissions through the dissemination of energy-efficient Improved Cookstoves (ICS) to rural households. The project displaces traditional three-stone fires and other inefficient biomass stoves, thereby reducing fuelwood consumption and associated emissions from the combustion of non-renewable biomass.

The project is implemented under a grouped project structure. During the second monitoring period (24-August-2023 to 12-September-2024), a total of 69,441 ICS units were distributed in two project instances, namely instance 1 located in Paderu division and instance 2 in Rampachodavaram division, both within the Alluri Sitarama Raju (ASR) district in the state of Andhra Pradesh, India. This includes 53,480 ICS installed in the newly added instance (instance 2) during the current monitoring period, 15,961 ICS in instance 1, and in MP1 total of 46,160 stoves were distributed bringing the cumulative total to 115,601 ICS disseminated under the project. Stove deployment and beneficiary details are recorded in the ICS distribution database /09/.

The project applies the approved VCS methodology VM0050: Energy Efficiency and Fuel Switch Measures in Cookstoves (Version 1.0) /B02/. The ICS units enhance thermal efficiency and reduce the consumption of non-renewable biomass, resulting in verifiable GHG emission reductions. For the second monitoring period, the project reports the following emissions data:

- Baseline emissions were estimated at 311,990 tCO₂e,
- Project emissions were calculated at 130,890 tCO₂e,
- Leakage emissions amounted to 9,055 tCO₂e,
- Resulting in net emission reductions of 172,045 tCO₂e.

The project proponent is Tramontana Asset Management, which retains full rights to the Verified Carbon Units (VCUs) /06/. The project has been implemented in accordance with the registered/revised VCS PD/03/, and the emission reductions /02/ have been calculated conservatively in line with the approved methodology.

In addition to its climate benefits, the project delivers co-benefits aligned with sustainable development goals (SDGs), including reduced indoor air pollution, improved health outcomes, time savings in fuel collection, and enhanced well-being of indigenous communities residing in remote and underserved areas of rural India.

2 VERIFICATION PROCESS

2.1 Method and Criteria

Key Milestones	Date
Kick off Meeting	21-January-2025
Desk Review	25-June-2025 to 30-June-2025

On-site Inspection	01-July-2025 to 04-July -2025 and 25-september-2025 to 26-september-2025
Opening Meeting	01-July -2025
Closing Meeting	04-July -2025
DVR Issuance	15-July-2025
Technical Review	03-October-2025

The verification was carried out in three distinct phases to ensure methodological rigor, data integrity, and compliance with the VCS framework:

1. Desk Review and Completeness Check

A thorough desk-based review was conducted of all relevant documentation to ensure compliance with the applicable standards and methodological requirements. This included the registered/revised VCS PD /03/, the monitoring report for the second monitoring period /01/ and the applied monitoring methodology /B02/ along with its associated tools. Particular attention was given to the monitoring plan, especially with regard to the frequency of measurements, the quality and calibration status of metering equipment, and the implementation of QA/QC procedures. The desk review also served to confirm the completeness, transparency, and consistency of the submitted data, as well as the accuracy and conservativeness of the emission reduction calculations and supporting evidence provided by the Project Proponent.

2. On-site Interviews and Data Verification

On-site interviews were conducted with project stakeholders and operational personnel to evaluate the on-ground implementation and monitoring of the project in alignment with the registered/revised PD. This phase included:

- Verification of the implementation and operation of the project against the validated PD,
- Review of data generation, aggregation, and reporting systems for key monitoring parameters,
- Examination of operational and data collection procedures to ensure adherence to the monitoring plan,
- Cross-checking reported monitoring data against independent sources (e.g., purchase and distribution records),
- Assessment of assumptions and parameters used in emission reduction calculations,
- Verification of the implementation of QA/QC procedures in accordance with both the methodology and the PD.

Follow-up interviews were conducted as necessary to clarify discrepancies and gather additional information.

3. Resolution and Final Reporting

Any outstanding issues or discrepancies identified during the verification process were addressed in consultation with the Project Proponent. Upon satisfactory resolution of all findings, the verification team finalized the verification conclusions and issued this report, along with the signed VCS Verification Deed of Representation.

2.2 Document Review

The verification was conducted as a document-based audit to assess the accuracy, completeness, and consistency of the information provided by the Project Proponent in accordance with the VCS Standard v4.7 /B01/ and the applied methodology VM0050 v1.0 /B02/. Carbon Check performed a detailed review of the registered/revised VCS PD/03/, the monitoring report for the second monitoring period version 1.3 /01/, and all relevant supporting documentation submitted by the Project Proponent.

This review included a cross-check of reported monitoring data against primary data sources such as the improved cookstove (ICS) distribution database, stove tracking records, and user monitoring survey outputs. The evaluation paid close attention to the implementation of the monitoring plan as outlined in the registered PD, specifically focusing on the appropriateness of measurement frequencies, quality and calibration of monitoring equipment, and the application of QA/QC procedures.

Additionally, the audit reviewed the emission reduction calculations, default values, and assumptions used in the ex-post spreadsheets to confirm consistency with the updated methodology requirements. Any inconsistencies or gaps identified during the initial review led to the issuance of Corrective Action Requests (CARs) and Clarification Requests (CLs), which were duly addressed by the Project Proponent. The revised monitoring report version 1.3 /01/ and associated evidence were subsequently reassessed to ensure the resolution of these findings. All reviewed documents and references have been documented in Appendix 2.1.

2.3 Interviews

As part of the verification process, an on-site visit to the project activity was conducted from 01-July-2025 to 04-July-2025. The purpose of the visit was to assess the implementation and ongoing operation of the project, verify the accuracy of the information provided in the Project Description (PD), Monitoring Report, and supporting documents, and to resolve any issues identified during the document review.

Structured interviews were conducted with key project personnel to confirm that the project was implemented in accordance with the registered PD/03/ and applicable methodology requirements. The interviews also served to verify that the monitoring systems, data collection processes, and emission reduction calculations were based on accurate and consistent information.

The on-site visit was done again from 25-september-2025 to 26-september-2025 to resolve one CAR.

The personnel interviewed and the main topics discussed are summarized below:

S. No	Date	Name	Organisation	Topic	Interviewed by
1.	01-July - 2025 to 04-July - 2025	Dr. Ramanan Loganathan	Tramontana	Interviews and field discussions were conducted to evaluate the project's performance against the approved PD, Monitoring Report, and applied methodology for both instances. Key topics included:	Manas Hadler, Temesgen Areaya and Nisha Jha
2.	01-July - 2025 to 04-July - 2025	Jessica Dring-Morris	Tramontana	- Project implementation and alignment with the PD in Instance 2. - ICS distribution status, operational performance, and usage patterns. - Emission reduction monitoring, including data collection, accuracy, and consistency with methodology.	
3.	01-July - 2025 to 04-July - 2025	Saibaba Abburi	Tramontana	Baseline scenario validation for Instance 2, supported by field data and KPTs.	
4.	01-July - 2025 to 04-July - 2025	Narasimha Barpata	Tramontana	- Monitoring framework, including data flow, documentation, and household-level verification. - Verification of monitored parameters, including sampling methods, KPT	

				data, and ER calculations as per VM0050. • Verification of SDG contributions, including reduced fuelwood use, health improvements, and time savings. • Assessment of “No Net Harm” compliance, with documented mitigation for identified risks. • Legal ownership and entitlement to carbon credits, verified through declarations and legal documentation. • Environmental and social co-benefits, confirmed through stakeholder interviews. • Stakeholder engagement and community participation, including meeting records and community feedback. • Grievance redress mechanism, reviewed for accessibility, transparency, and any recorded grievances. • Governance and safeguards, ensuring compliance with national regulations and project-specific commitments. • Verification of control households, confirming stability of baseline conditions and absence of leakage or project interference.	
5.	02-July - 2025	Vanthal Bangaru Naidu	Tramontana (Mandal Coordinator)	• Standard operating process • Baseline survey	

6.	03- July - 2025	Thodomu Mohan Rao	Tramontana (Mandal Coordinator)	• Distribution • Maintenance and grievance redressal • Monitoring survey • KPT • Data recording and QA/QC procedure • Training	
7.	04- July - 2025	Madivi Sundariah	Tramontana (Mandal Coordinator)		
8.	03- July - 2025	G. Johnmahesh	Tramontana (Mandal Coordinator)		
9.	03- July - 2025	Gonnafattula Uday Kumar	Tramontana (Manager)		
10.	03- July - 2025	Jabbadu Dhanalaksh mi	Tramontana (Computer Operator)	• Standard operating process • Data recording and QA/QC procedure • Training	
11.	03- July - 2025	A. Sai	Tramontana (Operation Analyst)		
12.	03- July - 2025	L. Dhanurajo	Tramontana (Maintenance Head)	• Standard operating process • Maintenance and grievance redressal • Data recording and QA/QC procedure • Training	
13.	09- July - 2025	Sourav Dey	Sustainergy Tech	• Product design • R&D • Technical parameters including efficiency • Supply agreement • Customer support • Training	
14.	09- July - 2025	Dr. Amit Ranjan Verma	Koshish Sustainable Solutions Pvt Ltd (Head – ESG & Carbon Markets)	• WBT demonstration • WBT protocol • Team competency • Accreditation • Training	
15.	02- July - 2025 to	Sadala Sundaramna	LSC attendee (Potential end user)	• Local stake holder consultation • Environmental and social co-benefits	

	03-July - 2025			- Stakeholder engagement and participation - Grievance redressal mechanism - Carbon financing	
16.	03-July - 2025 to 03-July - 2025	Miriyala Anil Kumar	LSC attendee (Potential end user)		
17.	04-July - 2025	Vaka Rajeswari	LSC attendee (Local Community Representative)		
18.	01-July - 2025	Majji Suramma	End user (Instance 1)	- Project participation - Functionality - Baseline/project survey - Baseline/project KPT - Environmental and social co-benefits - Stakeholder engagement and participation - Grievance redressal mechanism - Training	
19.	01-July - 2025	Gudiya Nageswara Rao	End user (Instance 1)		
20.	01-July - 2025	Gollori Prahladh	End user (Instance 1)		
21.	02-July - 2025	Vanthala Aso	End user (Instance 1)		
22.	02-July - 2025	Masadi Kundiramma	End user (Instance 1)		
23.	02-July - 2025	Masadi Nagaraju	End user (Instance 1)		
24.	02-July - 2025	Seedari Padma	End user (Instance 1)		
25.	02-July - 2025	Boyini Eswaramma	End user (Instance 1)		

26.	03- July - 2025	Korra Appanna	End user (Instance 1)		
27.	03- July - 2025	Majji Balaraju	End user (Instance 1)		
28.	03- July - 2025	Pangi Pentamma	End user (Instance 1)		
29.	02- July - 2025	Kelela Lacchi Reddy	End User (Instance 2)		
30.	02- July - 2025	Karanam Addama	End User (Instance 2)		
31.	02- July - 2025	Yadala Rajababu	End User (Instance 2)		
32.	02- July - 2025	Yadala Talupulamm a	End User (Instance 2)		
33.	02- July - 2025	Aragati Parvathi	End User (Instance 2)		
34.	02- July - 2025	Damala Kantha	End User (Instance 2)		
35.	02- July - 2025	Valasavada Reddy	End User (Instance 2)		
36.	03- July - 2025	Batta Neelaveni	End User (Instance 2)		
37.	03- July - 2025	Gomu Venkatlaksh ami	End User (Instance 2)		
38.	03- July - 2025	Gomu Kalyani	End User (Instance 2)		

39.	03-July - 2025	Nokku Ganga	End User (Instance 2)		
40.	25-september-2025	Murram Shara	End User (Instance 2)	- Project participation - Functionality - Project KPT - Monitoring survey - Environmental and social co-benefits - Grievance redressal mechanism - Training	
41.	25-september-2025	Madakam Durga	End User (Instance 2)		
42.	25-september-2025	Nupa Chukkamma	End User (Instance 2)		
43.	25-september-2025	Madivi Somaiah	End User (Instance 2)		
44.	25-september-2025	Madakam Naveena	End User (Instance 2)		
45.	25-september-2025	Tellam Rajamma	End User (Instance 2)		
46.	25-september-2025	Murram Veerraju	End User (Instance 2)		
47.	25-september-2025	Tellam Laxmi	End User (Instance 2)		
48.	26-september-2025	Shyamala Lakshmaiah	End User (Instance 2)		
49.	26-september	Karam Shuseela	End User (Instance 2)		

	mber-2025				
50.	26-septe mber-2025	Bogga Seethamma	End User (Instance 2)		
51.	01-July - 2025	Killo Banjo	End User (WBT)	- Project participation - WBT	
52.	02-July - 2025	Kimudu Ramarao	End User (WBT)		
53.	03-July - 2025	Regam Chilakamma	End User (WBT)		
54.	04-July - 2025	Gollori Purnima	End User (WBT)		
55.	04-July - 2025	Thokala Demudamm a	End User (WBT)		
56.	04-July - 2025	Vanthala Kesbo	End User (WBT)		
57.	04-July - 2025	Midiyam Kannamma	Control household (Instance 2)	- Baseline scenario - Baseline survey and KPT	
58.	04-July - 2025	Sode Subbaiah	Control household (Instance 2)		
59.	04-July - 2025	Kanithi Subhadra	Control household (Instance 2)		
60.	25-septe mber-2025	Kirasani neelamma	End user (Instance 1)	- Project participation - Functionality - Project KPT - Monitoring survey - Environmental and social co-benefits - Grievance redressal mechanism - Training	Manas Hadler, Temesgen Areaya and Nisha Jha
61.	25-septe mber-2025	Vanthala joythi	End user (Instance 1)		

62.	25-september-2025	Vechangi boidei	End user (Instance 1)		
63.	26-september-2025	Vanthala Chinnabai	End user (Instance 1)		
64.	25-september-2025	Vanthala Sandya Rani	End user (Instance 1)		
65.	26-september-2025	Shilpa Lakshmi	End user (Instance 1)		
66.	25-september-2025	Korra Amala	End user (Instance 1)		
67.	26-september-2025	Shilpa suseela	End user (Instance 1)		
68.	26-september-2025	Korra Appalamma	End user (Instance 1)		
69.	26-september-2025	Korra Neelanna	End user (Instance 1)		
70.	26-september-2025	Pangi Parvathamma	End user (Instance 1)		

For the project activity, a total of 11 sample households per instance were selected from the monitoring population to verify the implementation and performance of improved cookstoves (ICS) in accordance with the applied methodology VM0050 v1.0 /B02/. These households were chosen to ensure representative coverage of stove usage, user behaviour, and adherence to the project's monitoring requirements.

As required under VM0050 v1.0 /B02/ on-site inspection was conducted at all 11 sample households per instance to assess the applicability of the methodology and verify compliance with key conditions, including the correct installation of ICS, user training, stove usage patterns, and maintenance practices. These interviews formed a critical part of verifying that the monitored emission reductions were based on accurate and credible data, and that the project continues to meet the criteria established in the registered Project Description and methodology.

2.4 Site Visits

A site visit was conducted by the verification team from 01-July-2025 to 04-July-2025, in accordance with the requirements of the VCS Standard v4.7 /B01/ and the applied methodology VM0050 v1.0 /B02/. The purpose of the visit was to independently verify the physical implementation of the project activity, assess monitoring practices, and confirm adherence to the registered/revised PD/03/ and applicable VCS requirements.

The verification team conducted field inspections in both existing and newly added project instance areas, with specific emphasis on the newly introduced distribution area in the Rampachodavaram division, Alluri Sitarama Raju district, Andhra Pradesh. A total of 11¹ households per instance were selected for on-site visits based on a risk-based and representative sampling approach to capture variation across geographic and demographic groups.

The on-site verification activities included:

- Confirmation of stove distribution and installation in sampled households to ensure consistency with the monitoring database;
- Validation of the baseline scenario and project boundary assumptions through direct observation and stakeholder discussions;
- Evaluation of data collection and monitoring procedures, including implementation of QA/QC systems as outlined in the registered monitoring plan;
- Direct interviews with end-users and project implementation personnel to assess stove usage patterns and adherence to operational protocols;
- Physical inspection of ICS units for condition, operational use, and presence of required identification and tracking elements;

¹ 14 households from instance 2 were selected including 3 control households

- Review of supporting documentation and records (e.g., distribution logs, training records) to triangulate field findings and assess data integrity.

Sampling Approach

During the current monitoring period (24-August-2023 to 12-September-2024), a total of 69,441 Improved Cookstoves (ICS) were distributed across two project instances in the Alluri Sitarama Raju (ASR) district of Andhra Pradesh, India—15,961 units in Instance 1 (Paderu division), 53,480 units in Instance 2 (Rampachodavaram division) and in MP1 total of 46,160 stoves were distributed with these additions, the project has cumulatively disseminated 115,601 ICS units to households under the activity.

The sampling plan applied by the Project Proponent (PP) follows the approved monitoring methodology VCS Methodology VM0050, Version 1.0 /B02/, the revised Project Design Document (PD) /O1/, and the CDM Standard: Sampling and Surveys, Version 09.0 /B04/.

The PP employed Simple Random Sampling in line with VM0050 and the registered monitoring plan. “Sampling activities—including household surveys and Kitchen Performance Tests (KPTs) were carried out by trained field teams using the Tramontana ICS software. Each stove carried a unique serial number, ensuring traceability.

The following parameters are not calculated on a sample basis, but instead on the total project population and do not form part of sample-based monitoring:

- $N_{j,k,y}$: Number of commissioned devices of type j from batch k in year y
- n_{HH} : Number of ICS distributed to each household

For the above parameters VVB randomly selected 96 samples from the database and performed full data check at higher data aggregation levels and sampling at original data levels (based on 95/10 confidence / precision and 50 % response distribution²)

The following monitoring parameters were assessed through surveys and testing:

- $\eta_{j,k,y}$: Proportion of ICS still regularly used
- $BC_{b,i,y}$: Average baseline fuel consumption
- $BC_{p,j,k,y}$: Average project fuel consumption
- $Hh_i / Hh_{j,k}$: Household size under baseline/project conditions
- $\eta_{new,avg,y}$: Weighted average efficiency of project stoves (via WBT)

² <http://www.raosoft.com/samplesize.html>

For parameters $\eta_{\text{new,avg},y}$, the values were already determined using a separate approach as per the methodology. The methodology required efficiency tests to be conducted for those stoves which have been used for (at least) one year ($y=1$). Based on this WBTs were undertaken on September 2024 on 18 samples (6 ICS with 3 WBTs conducted for each). VVB has verified the efficiency of stoves for $y=1$ by taking all 6 PP samples using Simple Random Sampling.

For the other parameters namely, $BC_{b,i,y}$, $BC_{p,j,k,y}$, and Hh_i / $Hh_{j,k}$, the PP conducted baseline survey for instance 2 on 1472 households. Additionally, for both instances a total of 294 KPTs have been conducted. Specifically, 195 project KPTs were conducted as paired samples (with baseline KPTs), and 99 KPTs were conducted. VVB applied the CDM Sampling and Surveys for CDM Project Activities and Programmes of Activities, Version 09 /B04/ with an acceptance sampling plan using an Acceptance Quality Limit (AQL) of 0.5%, Upper Quality Limit (UQL) of 20%, producer/consumer risks of 10% each, and an acceptance number (c) of 0. A total of 11 samples were selected, and oversampling was conducted on baseline by adding 3 additional samples, reaching 14 in total.

Parameter	Verification approach	Population (for VVB's sample)	VVB's Sample Size
$N_{j,k,y}$ Number of commissioned devices of type j from batch k in year y	COM	Instance 1: 62,121 Instance 2: 53,480	Instance 1: 96 Instance 2: 96
n_{HH} Number of ICS distributed to each household			
$n_{j,k,y}$ Proportion of commissioned devices of type j from batch k still regularly used in year y	ASP	Instance 1: 673 Instance 2: 515	Instance 1: 96 Instance 2: 96
Hh_i $Hh_{j,k}$ Average household size of the target population using baseline device type i Average household size of the target population using project device type j from batch k	ASP	Instance 1: 151 Instance 2: 95	Instance 1: 11 Instance 2: 11
n_{HH} Number of ICS distributed to each household	COM	Instance 1: 62,121 Instance 2: 53,480	Instance 1: 96 Instance 2: 96

BC_{b,i,y} Average fuel quantity per baseline device type i in year y	ASP	Instance 1: 151 Instance 2: 154	Instance 1: 11 Instance 2: 11
BC_{p,j,k,y} Average quantity of fuel used by project device type j from batch k during year y	ASP	Instance 1: 150 Instance 2: 144	Instance 1: 11 Instance 2: 11
η_{new,avg,y} Weighted average efficiency of project devices in year y	ASP	6	6

Compliance with Sampling Standard /B04/

The VVB confirms that the PP's sampling design, implementation, and reported monitoring results were consistent with the requirements of **Paragraph 33 of the CDM Standard: Sampling and Surveys, Version 09.0 /B04/**. The VVB further confirms adherence to Paragraphs 25–28, which require the DOE to independently verify that the PP's sampling is methodologically sound, applies appropriate verification sampling, and achieves the required **90/10 confidence/precision level**.

All monitored values were supported by verifiable evidence and field checks. The VVB found **no deviations** in the PP's sampling and monitoring implementation.

Additionally, the verification team interviewed project implementation personnel and field agents to assess operational consistency, training adequacy, and documentation procedures. All findings and observations from the field visit were documented, and follow-up clarifications were sought from the Project Proponent where minor issues were identified.

This on-site assessment significantly contributed to the verification team's confidence in the credibility of the monitoring systems and the integrity of the reported emission reductions for the monitoring period 24-August-2023 to 12-September-2024. Based on the site visit findings and supplementary evidence, the verification was finalized with a reasonable level of assurance, confirming the project's conformance with applicable VCS criteria.

Activity	Place	Topics
Opening meeting	Project Site	Brief description of project activity and scope of audit.

Interviews with representatives of PP including distribution/installation, implementation and monitoring/testing teams	Project Site / PP's Local Office in Andhra Pradesh	• Implementation status • ICS distribution • QA/QC process • Database management • Trainings conducted • On-going stakeholder consultation • Grievance redressal • Carbon credits ownership rights • Project documentation • Emission reduction calculations • Supporting evidence
Interview with ICS manufacturing partner	Online	• Product design • R&D • Technical parameters including efficiency • Supply agreement • Customer support • Training
Interview with third-party testing agency responsible for conducting water boiling test (WBT)	Online	• WBT demonstration • WBT protocol • Team competency • Accreditation • Training
Interviews with LSC attendees	Project Site / PP's Local Office in Andhra Pradesh	• Local stake holder consultation • Environmental and social co-benefits • Stakeholder engagement and community participation • Grievance redressal mechanism • Carbon financing
Interviews with end-users	Project Site	• Cross-check PP's monitoring survey records • SDG benefits • Unique Identification of the ICS • Carbon credits waiver agreement

		ICS warranty
Closing meeting	Project Site	Discussion on findings

2.5 Resolution of Findings

This section presents the outcomes of the verification process, including the identification and resolution of findings resulting from the review of project documentation, field verification, and interviews with project personnel and stakeholders.

During the course of this verification, the verification team raised a total of 09 Corrective Action Requests (CARs) and 05 Clarification Requests (CLs). 02 Forward Action Requests (FARs) were raised during this monitoring period, and there were 02 outstanding FARs from previous verifications.

Findings were issued in response to identified discrepancies, gaps in supporting evidence, or areas of insufficient clarity with respect to the applied methodology (VM0050 v1.0) /B02/, the registered VCS Project Description /03/, and the VCS Standard v4.7 /B01/. Each finding was formally communicated to the Project Proponent (PP) and tracked through to closure.

Verification Finding Categories

- Corrective Action Request (CAR):** Raised when a material discrepancy was identified that could impact the accuracy of reported emission reductions or when specific methodological or VCS requirements were not met.
- Clarification Request (CL):** Raised where additional information or clarification was required to establish compliance or improve transparency.

Each CAR and CL was formally communicated to the Project Proponent. The proponent responded with the necessary documentation, clarifications, and updates to the monitoring report and supporting spreadsheets where applicable. The verification team reviewed these responses and determined that the actions taken were sufficient to close the findings.

The resolution of the CAR led to minor revisions in the monitoring documentation, specifically ensuring the consistency and completeness of parameter reporting. CLs mainly addressed the transparency of monitoring procedures, stove usage documentation, and data traceability. No changes to the estimated emission reductions were required as a result of the findings.

A detailed summary of each CAR and CL, along with the issues raised, responses provided by the Project Proponent, and the conclusions drawn by the verification team, is presented in Appendix 5.

2.5.1 Forward Action Requests

Forward Action Request (FAR) is to be raised when the monitoring and reporting require attention and/or adjustment for the next verification period. FARs do not relate to VCS requirements for issuance of ERs achieved during subject monitoring.

The verification team has raised 2 FAR during this verification.

2.6 Eligibility for Validation Activities

The project activity falls under sectoral scope 03: Energy Demand and is categorized as a Type I - Energy Efficiency Improvement Project. It specifically involves the replacement of non-renewable biomass-fired baseline cookstoves with more efficient project devices that utilize the same fuel type. The intervention is implemented as a grouped project activity under the VCS Program, in accordance with the requirements of methodology VM0050, Version 1.0 /B02/.

Carbon Check is accredited for both validation and verification of projects under this sectoral scope. Therefore, the eligibility requirements set out in the VCS Program Guide /B01-b/ for validation and verification activities are fully met.

3 VALIDATION FINDINGS

3.1 Methodology Deviations

There is no methodology deviation identified during the current monitoring period

3.2 Project Description Deviations

N/A

3.3 New Project Activity Instances in Grouped Projects

A new project activity instance (Instance 2) was included during the second monitoring period, involving the distribution of 53,480 improved cookstoves. The VVB has assessed the eligibility of Instance 2 against the criteria outlined in the registered project description and the requirements of the VCS Standard v4.7.

Eligibility criteria	Assessment and conclusion
Meet the applicability conditions set out in the methodology applied to the project	Instance 2 meets all applicability conditions as per the applied methodology. This has been confirmed through review of Section 3.2 of the PD /03/. The VVB concludes that the criterion is met.
Use the technologies or measures specified in the project description	The instance employs improved cookstoves (ICS) consistent with the technologies described in the PD /03/. No deviation in type, model, or application was identified. This criterion is satisfied.
Apply the technologies or measures in the same manner as specified in the project description	The ICS were implemented exactly as specified in the PD, with no alternative technologies or deployment methods used. The VVB confirms conformance with this requirement.
Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area	A region-specific baseline survey was conducted for Instance 2. The data and assumptions align with those presented in Section 3.4 of the PD /03/ and the applied methodology. For the new instance, the project location was checked with KML file /28/ and VVB confirmed it is within the project boundary. Criterion is fulfilled.
Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area	Additionality is established through demonstration of: (a) Regulatory Surplus — no legal mandate for the project, and (b) Positive List eligibility per the methodology. The “Project Method” step is not applicable. VVB confirms this criterion is met.

In accordance with Section 3.6.6 of the VCS Standard v4.7 /B01/, any new project activity instances added after validation must meet a defined set of additional eligibility criteria. The VVB has assessed Instance 2 against these specific requirements to ensure its conformance with the program rules for grouped projects. The assessment is based on submitted documentation, databases, stakeholder records, baseline and emissions reduction calculations, and contractual evidence provided by the project proponent.

Eligibility criteria	Assessment and conclusion
Occur within one of the designated geographic areas specified in the project description	Instance 2 is located in Rampachodavaram division, ASR district, Andhra Pradesh, within the geographic boundary of India as specified in Section 1.8 of the PD/03/. The location complies with the designated areas.
Conform with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial conformance with multiple sets of eligibility criteria is insufficient	Instance 2 complies with the full set of eligibility criteria for inclusion under the existing project activity as described in the table above. As evidenced in table above, Instance 2 fully meets a complete set of eligibility criteria for inclusion. Partial conformance is not applicable
Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate conformance with the applicable set of eligibility criteria and enable evidence gathering by the validation/verification body	The monitoring report /01/ includes comprehensive documentation: baseline and distribution databases, stakeholder meeting minutes, and methodology calculations. The VVB confirms sufficiency of submitted evidence.
Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions)	Ownership of emission reductions is clearly assigned to the Project Proponent through signed license agreements (digital and/or thumbprint). Agreements are documented in the Tramontana ICS software and stored as signed PDFs.
Have a start date that is the same as or later than the grouped project start date	Instance 2 distribution began on the 7-January-2024, which is after the start date of the grouped project of 21-January-2023.
Only be eligible for crediting from the later of start date of the project activity instance or the start of the verification period in which they were added to the grouped project, through to the end of the total project crediting period	The Instance 2 start date 7-January-2024, which is after the start of the current verification period (24-August-2023), and is thus compliant with the VCS Program rules.

Not be or have been enrolled in another VCS project	Instance 2 is not and will not be enrolled in any other VCS project. A declaration of exclusivity has been provided.
Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 – 3.6.9 of the VCS Standard v4.7	Calculations provided in the Emissions Reductions spreadsheet confirm that Instance 2 does not exceed the 60 GWh/year energy savings threshold. CDM AMS-II.G guidelines are followed.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☒ Yes ☐ No

The VVB has reviewed the baseline reassessment conducted for the project during the second monitoring period, specifically for **Instance 2**, while noting that the baseline for **Instance 1** remains unchanged and valid. The reassessment was performed to account for the commencement of ICS distribution in a new project area, which, although sharing comparable characteristics with Instance 1, required a separate baseline survey to ensure an accurate representation of local practices.

Applicability of Methodology

The reassessment applied the latest approved version of the methodology **VM0050 v1.0 /B02/**, including requirements introduced in October 2024. Specifically, additional baseline surveys and Kitchen Performance Tests (KPTs) were conducted to incorporate the new methodology questions regarding daily and weekend meal preparation. The VVB confirms that the methodology is applicable and appropriately implemented for both project instances.

Quality and Completeness of Data and Assumptions

- **Survey Coverage:** Baseline surveys were conducted across 11 mandals, covering 100 randomly selected villages, with 1,472 household surveys completed. Oversampling was applied to enhance data reliability.
- **Stove Usage:** 70% of households used traditional fixed mud clay stoves; 30% used three-stone stoves. 100% of households used stoves for cooking/tea/water boiling, and 79% for heating.
- **Fuel Consumption:** KPTs measured **16.55 kg/household/day** in Instance 2, compared to **13.22 kg/household/day** in Instance 1. Differences are justified by higher forest density and lower restrictions on wood collection in Instance 2.

- **Household Characteristics:** Average household size was **3.13 equivalent standard male adults (~4 persons)**, consistent with regional statistics. Women were the primary stove users and fuel collectors.
- **Socioeconomic Context:** Average family income USD 1.23/day; >92% engaged in agriculture; all households hold BPL/ration cards.

The VVB concludes that the evidence and assumptions used to determine the baseline scenario are **complete, consistent, and robust**.

Assessment of Updated Project Description

The updated baseline survey results for Instance 2, including stove usage, fuel consumption, and household characteristics, have been properly integrated into the project description. No changes were required for Instance 1, confirming that the **baseline scenario remains valid** for that area. All data were collected and documented in accordance with the PD /03/specifications and VM0050 v1.0 /B02/ methodology requirements.

Consideration of Policies and External Circumstances

The reassessment includes an evaluation of relevant national and sectoral policies, including the PMUY LPG program. The VVB notes that such policies have minimal impact in the Instance 2 area due to remoteness and low program uptake. Consequently, there is no material effect on the validity of the baseline scenario.

Appropriateness of Revised Baseline Estimates

The revised baseline emissions for Instance 2 reflect measured fuel consumption and usage patterns. The estimates are consistent with observed regional practices, the availability of firewood, and socio-economic conditions. Ex-ante projections for future baseline periods are considered **appropriate and realistic**.

The reassessment effectively documented socioeconomic conditions that reinforce the persistence of baseline practices. Notable issues include: firewood scarcity, negative health effects from indoor smoke, physical burdens of fuel collection, and lack of awareness of improved technologies. These observations further substantiate the appropriateness of the chosen baseline.

The VVB reviewed the updated baseline scenario of **Instance 2**, while confirming that the baseline scenario for **Instance 1** remains unchanged and valid. The reassessment was required due to the expansion of the project to a new area and was conducted using the **latest approved methodology VM0050 v1.0 /B02/**, including updates introduced in October 2024.

The VVB verified that additional baseline household surveys and Kitchen Performance Tests (KPTs) were conducted in accordance with the new methodology requirements, including questionnaire in Appendix 3 of the applied methodology during the survey. The survey design, sample size, and selection process for the participating households were reviewed and found to be adequate and representative. Data collection procedures and assumptions used for stove usage, fuel consumption, and household characteristics were assessed for completeness, consistency, and plausibility.

As no recent government statistics have been published for the project area, the consumption results have instead been compared to an independent study conducted in the project area between December 2024 and January 2025 by an independent specialist local company - Koshish Sustainable Solutions. The VVB confirmed that **recent and credible data sources** were used and survey data were reviewed for adequate geographic coverage, randomization, and sample size.

Therefore, the updated baseline scenario has been assessed and established in compliance with VM0050 v1.0 /B02/ and that the resulting baseline is supported by recent and credible data. Based on the review of data, surveys, methodology application, and contextual factors, the VVB concludes that:

- The baseline reassessment for Instance 2 has been conducted in accordance with VM0050 v1.0 /B02/ requirements.
- The resulting baseline scenario is accurate, realistic, and valid.
- The original baseline for Instance 1 remains unchanged and valid.
- Updated baseline emissions estimates are robust and suitable for monitoring and crediting purposes.
- The baseline is supported by recent, reliable, and credible data

4 VERIFICATION FINDINGS

4.1 Project Details

Item	Evidence gathering activities, evidence checked, and assessment conclusion:				
	Audit Type	Period	Program	VVB Name	Number of Years
Audit history					

	Joint Validation + Verification	21-January-2023 to 23-August-2023	VCS	Earthood Services Private Limited	0.586
	Requantification + Methodology Change	21-January-2023 to 23-August-2023	VCS	Carbon Check (India) Private Limited	0.586
	Verification	24-August-2023 – 12-September-2024	VCS	Carbon Check (India) Private Limited	1.058
Double counting and participation under other GHG programs The monitoring system is implemented and fully functional to generate emission reductions without any double counting. The project is not receiving or seeking credit for GHG emission reductions or removals from the same project activity under any other GHG program. The verification team, through document review and on-site interviews, confirms that the emission reductions are unique and not claimed elsewhere, in conformance with the requirements of §3.23 of the VCS Standard Version 4.7 /B01/. The Project Proponent has provided all required information confirming that the project is not registered or seeking registration under any other GHG program, apart from its registration with Verra. The Project Proponent has also confirmed that the project has not been rejected by any other GHG program to date. The VVB conducted a comprehensive review of the Verra registry and other relevant carbon registries to identify any existing or overlapping carbon projects in the project region. This review found no evidence of any other carbon projects or schemes operating in the area. Consequently, the risk that two stoves within the same household could belong to different carbon projects or schemes has been eliminated. It is noteworthy that the project activity is registered under the Verra Verified Carbon Standard (VCS) with registration number 4457. This registration reflects the project's exclusive association with the VCS					

	Program and ensures that no double issuance or participation under other mechanisms has occurred. The validation and verification body confirms, based on the review of all evidence provided, that there is no risk of double counting or double issuance for the reported GHG emission reductions during the monitoring period 24-August-2023 to 12-September-2024.
No double claiming with emissions trading programs or binding emission limits	The assessment has been conducted in conformance with the requirements under §3.24.3 of the VCS Standard Version 4.7 /B01/ and confirms the absence of any potential double claiming of GHG emission reductions. VVB reviewed each ICS is assigned a unique Identifier in the form of a serial number, displayed on the ICS as both a QR code and in written form. This identifier is linked to the individual household through a proprietary ICS software application, which records all relevant beneficiary and stove data. The VVB validated this system by cross-checking the online distribution database, ensuring that each stove could be traced to a specific household. Additionally, Section 5.2 of the PD explains how the project accounts for rare cases where a household receives more than one stove. The parameter n_{HH}, representing the number of stoves per household, is applied in emission reduction calculations. The VVB verified the correct use of this parameter for the monitoring period (see “ICS-01 Emission Reductions M2”), ensuring that emissions from multiple stoves within the same household are not double counted. VVB confirms that this project is not registered or seeking registration in the GCC, GS, or CDM project databases, according to VVB's investigation of: GS registry (https://registry.goldstandard.org/projects?q=&page=1); GCC project database (https://projects.globalcarboncouncil.com/pages/submitted_projects) ; and CDM project database (https://cdm.unfccc.int/Projects/projsearch.html)

No double claiming with other forms of environmental credit	The project activity has not sought, received, nor is it planning to receive credit from another GHG-related environmental credit system. The verification team, through a detailed review of project documentation and on-site interviews with the Project Proponent, confirms that no other forms of environmental crediting—such as renewable energy certificates, biodiversity credits, or water benefit credits—have been applied for or awarded in relation to the emission reductions generated by this project. This conclusion is made in conformance with §3.24.1 of the VCS Standard Version 4.7 /B01/, ensuring the integrity and uniqueness of the verified GHG emission reductions.
Supply chain (scope 3) emissions double claiming	The GHG emission reductions or removals are not associated with any supply chain emissions, as the project proponent or authorized representative is neither a buyer nor a seller of a product whose emissions footprint is affected by the project activity. Therefore, the verification team, by means of document review and on-site interviews, confirms that there is no risk of Scope 3 emission double claiming for this project activity, in conformance with §3.24.7 of VCS Standard Version 4.7 /B01/.
Sustainable development contributions	The project has implemented the activities that result in the Sustainable Development (SD) contributions described in Section 1.12 of the monitoring report. These activities have been assessed and verified through document review and stakeholder consultation, confirming alignment with the declared SD goals.
Additional information relevant to the project	No commercially sensitive information has been excluded from the public versions of the project documents beyond what is permitted under the VCS Program.

4.2 Safeguards and Stakeholder Engagement

4.2.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
------	--

Stakeholder identification	The verification team through review of MR /01/, VCSPD /03/ and interviews with representatives of PP confirms that sufficient mechanism is in place to identify stakeholder and PP has identified relevant stakeholders as described in the MR /01/.
Legal or customary tenure/access rights	The project maintains close engagement with local communities to address any concerns related to tenure or access rights. To date, no conflicts have been reported. The VVB deems this acceptable based on desk review and interviews conducted on-site visit.
Stakeholder diversity and changes over time	The project primarily engages indigenous and socio-economically disadvantaged communities. The composition and diversity of stakeholders have remained consistent since validation. The VVB finds this acceptable through document review and on-site stakeholder engagement.
Expected changes in well-being	The anticipated positive impacts on stakeholder well-being, including reduced drudgery and improved safety, are consistent with those reported during validation. No negative changes are expected. This conclusion is supported by evidence reviewed by the VVB.
Location of stakeholders	The geographic distribution of stakeholders remains unchanged. Beneficiaries continue to be located within the Paderu division, ASR district, Andhra Pradesh, India. The VVB confirms this based on project records and interviews/19/
Location of resources	The project does not interfere with stakeholders' land or resource access, which remains consistent with previous assessments. The VVB accepts this based on desk review and stakeholder consultation.

4.2.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
------	--

Ongoing consultation	Ongoing consultation is actively maintained with local communities through the employment of 155 local people (51 full-time, 104 part-time) who ensure continued ICS usage and community engagement. The VVB reviewed the grievance mechanisms, including access to a local mandal representative and a dedicated phone number, which accommodate literacy challenges. The grievance register/13/ is regularly reviewed by the Project Proponent, and issues are addressed accordingly. Regular field visits and ICS performance monitoring are conducted and recorded digitally, providing a robust monitoring framework. The VVB confirms these arrangements meet VCS Program requirements based on document review and remote interviews.
Date(s) of stakeholder consultation	Stakeholder consultations for the second project instance in 11 mandals of Rampachodavaram division, ASR district, Andhra Pradesh, occurred from 8 th to 28 th November 2023. One meeting was held per mandal with a total attendance of 4,550 community members. The VVB reviewed minutes, attendance sheets, and summary schedules (provided in Appendix 1), confirming comprehensive and uniform consultation in conformance with VCS Program requirements.
Communication of monitored results	Monitored results were communicated effectively through local coordinators. For example, feedback from monitoring surveys about ICS maintenance was shared with beneficiaries, leading to reminders about ICS care. The VVB considers the communication process to be transparent and effective, based on documentation and interviews.
Consultation records	Records including detailed minutes, attendance sheets, stakeholder questions, responses, and photographs were reviewed/25/. All documents are maintained in English and original languages. The VVB finds the record-keeping to be thorough and compliant with VCS Program standards.
Stakeholder input	All stakeholder questions and responses during meetings are documented and provided to the VVB and Verra. The project team's availability for further discussion after meetings ensures stakeholder satisfaction. No modifications to the project design were required based on these consultations. The VVB concludes that stakeholder input is appropriately captured and considered in line with VCS Program expectations.

4.2.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Consent	The VVB reviewed stakeholder meeting records and the ICS beneficiary agreement template. During the stakeholder meetings, the benefits and safe use of the ICS were explained and demonstrated. The process allowed stakeholders to raise concerns or questions in an open format. Consent was obtained voluntarily at the time of ICS distribution, with beneficiaries signing a license agreement after receiving a detailed verbal explanation of the terms. The VVB concludes that the project has obtained free, prior, and informed consent in conformance with VCS Program FPIC requirements.
Outcome of FPIC discussion	Based on a review of pre-baseline survey data, stakeholder consultation minutes, and community feedback, all beneficiaries provided consent to participate in the project and accept the license agreement terms. No grievances, forced displacement, or encroachment on land were recorded. The VVB confirms that the process and outcomes of FPIC discussions are compliant and that project activities have been implemented with full and informed community participation.

4.2.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Grievance received and steps taken to resolve the grievance including the outcomes of the resolution	The VVB reviewed internal monitoring records, grievance logbook (repair logs), and beneficiary interviews to assess grievance handling. Four stoves were reported as damaged. These were repaired on 25-September-2024, 20-September-2024, 11-October-2024 and 7-November-2024 respectively. Interviews confirmed that beneficiaries continued using the stoves during the damage period, and thus no impact on GHG emission reductions occurred. The VVB concludes that the grievances were appropriately addressed and resolved in a timely manner, without adverse impact on emission reduction calculations.

Grievance redress procedure	The VVB reviewed the grievance redress mechanism, including community access via local mandal representatives and a dedicated phone number. All grievances are recorded in a local register /13/and reviewed regularly by the Project Proponent. The process is found to be accessible, responsive, and in conformance with the VCS Program requirements. The VVB concludes that the grievance procedure is appropriate and effectively implemented.
-----------------------------	--

4.2.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
No comments were received as part of the stakeholder consultation or outside of the public comment period.	No actions were required as no public comments were received.	The VVB reviewed the public comment log and confirmed that no comments were submitted during or after the public comment period. As no feedback was received, no changes to the project design were necessary. The VVB concludes that the project proponent remains in conformance with the VCS Program requirements regarding stakeholder input and public consultation.

4.2.6 Risks to Local Stakeholders and the Environment

4.2.6.1 Management Experience

The verification team assessed the management capacity of the Project Proponent and its implementation partners through a review of project documentation, organizational charts, training materials, and interviews conducted with senior management and field-level coordinators.

Roles and responsibilities for the monitoring, recording, and reporting of parameters related to GHG emission reductions were found to be clearly defined and consistently applied throughout the monitoring period. The Project Proponent has implemented an effective management structure that ensures oversight of field operations, quality control, and data integrity in line with the registered monitoring plan and VCS Program requirements.

Interviews with key personnel confirmed that staff involved in monitoring activities possess the required technical competencies and receive periodic training aligned with their responsibilities. Structured training programs/12/ have been developed and delivered, particularly for local mandal coordinators and field staff, to strengthen capacity in areas such as data collection protocols, stove usage tracking, and stakeholder engagement.

The local implementation partner, subsidiary of Tramontana Asset Management, demonstrated strong operational capabilities, especially in community mobilization, beneficiary coordination, and logistical execution. The Project Proponent retains overall managerial control and provides strategic direction, risk management, and performance oversight.

Based on the evidence reviewed and interviews conducted, the verification team concludes that the project's management system is robust, well-organized, and effectively supports the successful implementation and continuous monitoring of the project activity. The demonstrated managerial experience and operational structure are fully consistent with the requirements of the applied methodology and the VCS Program.

4.2.6.2 Risk Assessment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human induced risks to stakeholders' wellbeing	Based on monitoring reports, stakeholder feedback, and field data, the project poses no such risks. The Tramontana ICS enhances household health by reducing indoor air pollution and burn injuries. The VVB concludes that no risk has been identified and project design is acceptable.
Risks to stakeholder participation	The VVB reviewed engagement strategies, field records, and conducted interviews. Ongoing community support, regular ICS checks, and responsive technical assistance effectively manage this risk. The VVB finds mitigation measures appropriate.
Working conditions	Interviews and field documentation confirmed that the project does not alter traditional work practices. Improved stove efficiency enhances safety and indoor conditions. No risks identified.
Safety of women and girls	Stakeholder feedback and baseline data confirm the project improves safety for women and girls by reducing exposure to hazardous fuel collection and indoor smoke. The VVB finds no risk and acknowledges positive gender-related outcomes.

Safety of minority and marginalized groups, including children	The VVB verified that project activities reduce risks from baseline conditions, especially for vulnerable groups such as children and marginalized communities. The use of ICS with enclosed flames and community training during consultations minimizes accident risks. Based on review of stakeholder meeting minutes and the monitoring report, the VVB concludes that the residual risk is low and mitigation measures are effective.
Pollutants (air, noise, discharges to water, generation and release of hazardous materials and chemical pesticides and fertilizers)	The VVB confirms that the Tramontana ICS reduces indoor air pollution and does not introduce any new environmental risks such as noise, water discharge, or hazardous material use. This is validated through performance evaluations and stakeholder interviews. The VVB concludes that no such risks exist under the project activity.

4.2.7 Respect for Human Rights and Equity

4.2.7.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination	Staff training materials and grievance protocols were reviewed. Interviews with PP and employees confirmed awareness of anti-discrimination policies. No incidents were reported. The VVB concludes that the risk is minimal and effectively managed.
Sexual harassment	The VVB assessed staff training/12/ procedures and code of conduct documentation/26/. Project training includes strict prohibitions against sexual harassment, with procedures in place for immediate investigation and disciplinary action. No complaints were received. Controls are deemed sufficient by the VVB.
Gender equity in labor and work	The VVB reviewed project hiring records, HR policy, and interviewed field staff. The project emphasizes inclusive hiring practices, with a focus on improving female participation in the workforce, which is traditionally low in the region. Equal pay for equal work is practiced and verified through payroll records. The VVB finds that gender equity risks

	have been addressed and no concerns were identified during the monitoring period.
Forced labor	The VVB assessed recruitment and contract review and confirmed all labor is voluntary. Training emphasizes prohibition of forced labor. No violations were noted. Risk is appropriately mitigated.
Child labor	The VVB confirmed that the project enforces a strict minimum age requirement (18+), validated through identity verification and employment records and on-site interviews with representatives of PP ad employees. Training programs emphasize the prohibition of child labor. There have been no instances of underage employment. The VVB confirms that the project fully complies with child labor law /B09/and no risk has been identified.
Human trafficking	Staff background checks and training materials were reviewed. Policies regarding human trafficking are clearly communicated to all project staff. No indicators or allegations of human trafficking have been identified during the monitoring period. The VVB concludes that this risk is adequately mitigated and no risk has been observed.

4.2.7.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk identified	The VVB reviewed the monitoring report, stakeholder consultations, and project documentation. The project complies with international human rights standards, including the UN Declaration on the Rights of Indigenous Peoples and ILO Convention 169. Participation was voluntary, with no land encroachment or rights violations. No grievances were reported. The VVB concludes the project poses no human rights risks and conforms to applicable requirements.

4.2.7.3 Indigenous Peoples and Cultural Heritage

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
	The VVB reviewed the monitoring report, stakeholder consultation records, and project implementation protocols. All activities are

No risks identified	conducted with respect to local customs and cultural heritage. Stakeholder engagement occurs in the local dialect, facilitated by community-based team members. The VVB confirms that cultural heritage and the rights of Indigenous Peoples and local communities are respected, with no risks identified during the monitoring period.
---------------------	--

4.2.7.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risks identified	The project activity involves distribution of fuel-efficient ICS, and it does not require acquisition of property. The VVB assessed land use, stakeholder interviews, and project documentation and interviews with end-users. No encroachment, relocation, or interference with legal or customary tenure or property rights was identified. The VVB confirms full respect of property rights throughout the project period. No risks were observed or reported.

4.2.7.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Summary of the benefit sharing plan	The project activity involves distribution of fuel-efficient ICS to households and therefore does not impact property rights, usage, or resources. Hence not applicable.
Benefit sharing during the monitoring period	N/A

4.2.8 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	The VVB reviewed project documents, monitoring data, and stakeholder feedback. The ICS reduces firewood use, thereby lowering deforestation and positively impacting biodiversity. No risks identified.

Soil degradation and soil erosion	Reduced firewood demand contributes to less forest degradation, preserving soil cover and preventing erosion. Field reports support this outcome. No risks identified.
Water consumption and stress	Cleaner combustion reduces water needed for cleaning, and improved soil conditions enhance water retention. Verified through beneficiary feedback. No risks identified.

4.2.8.1 Rare, Threatened, and Endangered species

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Species or habitat	Based on project location data and environmental records, the VVB confirms the project is not located in or near habitats of rare, threatened, or endangered species. No risk identified. Fuel-efficient ICS is installed at the households, and are not located in, or adjacent to habitats for rare, threatened, or endangered species. This has been checked by the verification team during the onsite audit.
Areas needed for habitat connectivity	As per the reasons stated above, the project activity does not adversely impact areas needed for habitat connectivity during the monitoring period. No risk identified.
...	...

	Evidence gathering activities, evidence checked, and assessment conclusion
Habitats for rare, threatened, and endangered species	NA as per the reasons stated above.
Areas for habitat connectivity	NA as per the reasons stated above.

4.2.8.2 Introduction of Species

Not Applicable as this project involves distribution of fuel-efficient ICS to households. Therefore, it does not involve any planting or introduction of any new species.

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
N/A	N/A

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
NA	As stated above, this project involves distribution of fuel-efficient ICS to households. therefore, the project does not involve any activity or implementation measure that will cause any existing invasives to thrive. Neither does it involve the usage of any non-native species. Hence, not applicable.

4.2.8.3 Ecosystem conversion

Item	Evidence gathering activities and evidence checked
Ecosystem conversion	Not applicable as the project does not affect existing ecosystems

4.3 Accuracy of Reduction and Removal Calculations

The equations and choices provided in the methodology as well as all other methodological tools, are correctly quoted in the Monitoring report /01/. The emission reductions of the project activity are calculated using the formulae mentioned in the applied methodology; VM0050 version 1.0 /B02/. The verification team reviewed the emission reduction spread sheets (ER sheets) /02/ and checked all the formulae, concluding that they are correct and in accordance with the monitoring plan of the PD /03/ and the applied monitoring methodology.

According to applied methodology VM0050 (version 1.0) /B02/the emissions are calculated as below:

Baseline Emissions

Baseline emissions are calculated using Equation 1 from the VM0050 v1.0 methodology/B02/:

$$BE_y = \sum_{i,j,k} EC_{i,y} \times N_{j,k,y} \times n_{j,k,y} \times (EF_{b,i,CO2} \times f_{NRB,y} + EF_{b,i,nonCO2}) \quad \text{Equation (1)}$$

Where:

$$BE_y = \text{Baseline emissions during year y (t CO}_2\text{e)}$$

$EC_{i,y}$	=	Average energy consumption of baseline device type i in year y (TJ)
$N_{j,k,y}$	=	Number of commissioned project devices of type j from batch k in year y
$n_{j,k,y}$	=	Proportion of commissioned project devices of type j from batch k that remain operating in year y (fraction)
EF_{b,i,CO_2}	=	CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO ₂ /TJ)
$f_{NRB,y}$	=	Fraction of woody biomass that is established to be non-renewable used by baseline device in year y (fraction)
$EF_{b,i,nonCO_2}$	=	Non-CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO ₂ e/TJ)
i	=	Baseline device type and its respective fuel type
j	=	Project device type and its respective fuel type

Following the VM0050 v1.0 methodology, the average energy consumption of baseline devices is calculated using:

$$EC_{i,y} = BC_{b,i,y} \times NCV_{b,i} \quad \text{Equation (2)}$$

Where:

$BC_{b,i,y}$	=	Fuel used per baseline device type i during year y (tonnes)
$NCV_{b,i}$	=	Net calorific value of baseline fuel for baseline device type i (TJ/tonne)

For Instance, 1, the following parameters are used:

Parameter	Value	Source
$BC_{b,i,y}$	4.83 tonnes	Baseline KPT Survey – Instance 1
$NCV_{b,i}$	0.0156 TJ/tonne	Methodology

$$EC_{i,0} = 4.83 \times 0.0156$$

$$= 0.0753 \text{ TJ}$$

Calculations for Instance 1 at $y = 0$ are provided in section 4.1 of the PD. From the PD, $EC_{i,0}$ for Instance 1 is calculated as 0.0753 TJ. As $BC_{b,i,y}$ is fixed for 5 years, the value of $EC_{i,y}$ for Instance 1 remains unchanged and therefore $EC_{i,1} = 0.0753 \text{ TJ}$.

For Instance, 2, the following parameters are used:

Parameter	Value	Source
$BC_{b,i,0}$	6.04 tonnes	Baseline KPT– Instance 2
$NCV_{b,i}$	0.0156 TJ/tonne	Methodology

$$EC_{i,0} = 6.04 \times 0.0156$$

$$= 0.0942 \text{ TJ}$$

$EC_{i,y}$ for both Instances must then be crosschecked to identify if there is any stove stacking. This is done by comparing the quantity of baseline energy consumption to energy used in the project scenario via a back-calculation with stove thermal efficiencies. If the results indicate a baseline consumption that is higher than that indicated through the back-calculation, the latter is applied as a conservative cap. This is calculated using the following equation:

$$EC_{est,y} = EC_{p,y} \times \frac{\eta_{new,avg,y}}{\eta_{old,avg}} \quad \text{Equation 3}$$

Where:

$EC_{est,y}$	=	Back-calculated energy consumption of the potential mix of devices and fuels in the baseline in year y (TJ)
$EC_{p,y}$	=	Energy used in project scenario by project devices during year y (TJ)
$\eta_{new,avg,y}$	=	Weighted average efficiency of project devices in year y (fraction)
$\eta_{old,avg}$	=	Weighted average efficiency of baseline devices that are replaced by project devices (fraction)

First $EC_{p,y}$ is calculated using:

$$EC_{p,y} = \sum_{j,k} BC_{p,j,k,y} \times NCV_{p,j} \quad \text{Equation 4}$$

Where:

$BC_{p,j,k,y}$ = Average quantity of fuel used by project device type j from batch k during year y (tonnes or m³)

$NCV_{p,j}$ = Net calorific value of project fuel used in project device type j (TJ/tonne or TJ/m³)

For this monitoring period, $EC_{est,y}$ is calculated for Instance 1 at $y = 1$ and Instance 2 at $y = 0$. We use the following parameters and apply Equation 4 to determine if a cap is applied:

Parameter	Value	Source
$BC_{p,j,1,1}$	1.66 tonnes	Project KPT- Instance 1
$BC_{p,j,2,0}$	1.82 tonnes	Project KPT- Instance 2
$NCV_{p,j}$	0.0156 TJ/tonne	Methodology
$\eta_{new,avg,0}$	36.01%	Water Boiling Test (see PD Appendix 2)
$\eta_{new,avg,1}$	34.73%	Water Boiling Test (see Appendix 2)
$\eta_{old,avg}$	15%	Methodology

For Instance 1:

$$EC_{p,1} = 1.66 \times 0.0156$$

$$= 0.0259 \text{ TJ}$$

$$EC_{est,1} = 0.0259 \times \frac{0.3473}{0.15}$$

$$= 0.0599 \text{ T}$$

As $0.0599 (EC_{est,1}) < 0.0753 (EC_{i,1})$, a cap is applied and $EC_{i,1}$ for Instance 1 is taken as 0.0599 TJ.

For Instance 2:

$$EC_{p,0} = 1.82 \times 0.0156$$

$$= 0.0283 \text{ TJ}$$

$$EC_{est,0} = 0.0283 \times \frac{0.3601}{0.15}$$

$$= 0.0680 \text{ TJ}$$

From above, $EC_{i,0}$ for Instance 2 is calculated as 0.0942 TJ.

As $0.0680 (EC_{est,0}) < 0.0942 (EC_{i,y})$, $EC_{i,0}$ for Instance 2 is taken as 0.0680 TJ.

To then calculate BE_y in Equation 2, $EC_{i,y}$ here is used along with the following parameter values for the second monitoring period:

Parameter	Value	Source
$BC_{b,1,y}$	4.83 tonnes	Baseline KPT Instance 1
$BC_{b,2,y}$	6.04 tonnes	Baseline KPT Instance 2
$EF_{b,i,CO2}$	112 tCO ₂ /TJ	Methodology
$EF_{b,i,nonCO2}$	9.46 tCO ₂ /TJ	Methodology
$NCV_{b,i}$	0.0156 TJ/tonne	Methodology
$f_{NRB,1}$	0.510	Methodology Tool (see PD section 4.4)
$f_{NRB,2}$	0.556	Methodology Tool (see section 5.4)

The calculation of BE_y for this monitoring period is presented in the sheet “ICS-01 Emission Reductions M2” and is used to determine ER_y as described in section 5.4 of this report.

Project Emissions

To calculate the project emissions, PE_y , for the second monitoring period, the approach described in section 4.2 of the PD is followed using the following input parameters:

Parameter	Value	Source
$EF_{p,i,CO2}$	112 tCO ₂ /TJ	Methodology

$EF_{p,i,nonCO2}$	9.46 tCO ₂ /TJ	Methodology
$NCV_{p,j}$	0.0156 TJ/tonne	Methodology
$BC_{pj,1,y}$	1.66 tonnes	Project KPT Instance 1
$BC_{pj,2,y}$	1.82 tonnes	Project KPT Instance 2
$f_{NRB,1}$	0.510	Methodology Tool (see PD section 4.4)
$f_{NRB,2}$	0.556	Methodology Tool (see section 5.4)

The calculation of PE_y for this monitoring period is presented in the sheet “ICS-01 Emission Reductions M2” and is used to determine ER_y as described in section 5.4 of this report.

Leakage Emissions: Leakage is accounted for using the adjustment factor of 0.95 in line with the methodology VM0050 version 1.0 /B02/ **Sampling approach:**

As assessed and verified in this section, the emission reductions claimed by the Project Participant (PP) for the project titled “TRAMONTANA ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities Across Rural India” for the current monitoring period (24 August 2023 to 12 September 2024) amount to 172,045 tCO₂e. The total number of improved cookstoves (ICS) deployed and operational during this period is 115,601 units.

The sampling plan implemented by the Project Proponent (PP) is in accordance with the applied approved monitoring methodology VM0050, version 1.0 /B02/ and the registered VCS Project Description /03/. The PP has appropriately applied a Simple Random Sampling (SRS) procedure, and the reliability levels were set at 90% confidence and 10% relative precision, in line with the applied methodology. As the VCS PD /03/ includes provisions for the use of simple random sampling, this approach is deemed acceptable by the verification team.

Sampling surveys have been conducted by trained field personnel /12/. The monitored parameters $N_{j,k,y}$, $n_{j,k,y}$, Hh_i , $Hh_{j,k}$, n_{HH} , $f_{NRB,y}$, $BC_{b,i,y}$, $\eta_{(new,avg,y)}$, and $BC_{p,j,k,y}$ are monitored through sampling-based surveys and standardized protocols. Monitoring of these parameters ensures compliance with the applied methodology VM0050, version 1.0 /B02/. The verification team has reviewed the survey records /08/, sample size calculations /10/, and associated documentation.

Parameter	Description	Monitoring frequency	Means of assessment by VVB
-----------	-------------	----------------------	----------------------------

$N_{j,k,y}$	Number of commissioned project devices of type j from batch k in year y	At the time of commissioning/distribution of ICS	The VVB verified the commissioning of project devices via the Tramontana ICS distribution database/09/, confirming the number and type of devices deployed. A total of 115,601 ICS units were distributed, as verified through distribution records and signed delivery receipts. The distribution records were cross-checked with the implementation plan to ensure alignment with reported batches.
$n_{j,k,y}$	Proportion of commissioned project devices of type j from batch k that are still being used regularly in year y	Expected annually	Based on the monitoring survey, the VVB confirmed active usage through household interviews and direct observations/19/. Spot checks and field records showed that ICS units were operational, with positive user feedback and no substantial breakdowns reported in the grievance logs maintained by the PP/13/.
n_{HH}	Number of ICS distributed to each household	At the time of commissioning/distribution of ICS	VVB confirmed that each household received one ICS unit as reflected in the distribution database/09/. Identity verification documents such as Aadhaar and BPL cards were reviewed to ensure no duplication or misreporting. The distribution count of 115,601 ICS units was confirmed to match the number of recipient households.
$f_{NRB,y}$	Fraction of woody biomass that is established to be non-renewable used by device in year y	A new value is determined every time the project activity instance is undertaken in a new state	The VVB reviewed state-specific NRB values based on peer-reviewed literature and/or government publications submitted in the revised PD report. In addition, the fNRB Excel sheet /20/ was assessed to validate consistency with referenced sources. For each new implementation region, the NRB factor was found to be justified and compliant with VM0050 v1.0/B02/ and VCS Program rules/B01/.

$BC_{b,i,y}$	Average quantity of fuel per baseline device type i during year y	Expected annually and will occur at least biennially	The VVB assessed baseline fuel consumption through structured monitoring surveys. WBT/KPT records were sampled and analyzed to calculate average fuel use for traditional stoves. These findings were cross-verified with kitchen observations and user recall data collected during field verification activities.
$BC_{p,j,k,y}$	Average quantity of fuel used by project device type j from batch k during year y	Expected annually and will occur at least biennially	The VVB confirmed project fuel consumption by evaluating monitored data sets and KPT results collected from sampled ICS users. Stove usage logs, user interviews, and observational data were reviewed to ensure methodological consistency with VM0050 v1.0 and the approved monitoring plan.
$Hh_i / Hh_{j,k}$	The average household size of the target population using baseline device type i and project device type j from batch k .	A new value is determined every time the project activity is undertaken in a new instance	Household size data was obtained through statistically representative surveys and validated against regional census benchmarks. Sampling protocols and calculations were reviewed and found to be in alignment with methodological specifications under VM0050 v1.0.
$\eta_{new,avg,y}$	Weighted average efficiency of project devices in year y	Expected Annually	The VVB reviewed efficiency test reports conducted in accordance with the Water Boiling Test (WBT) protocol. The weighted average thermal efficiency was confirmed from manufacturer specifications/18/ and independent laboratory results/24/. The findings were consistent with the required performance criteria under VM0050 v1.0.

The parameter $n_{j,k,y}$, representing the proportion of commissioned project devices in operation during the year, was determined through field monitoring surveys. The minimum required sample size was calculated in line with the Guideline: Sampling and surveys for CDM project activities and PoAs (Appendix 1, para 12), assuming a population size of 115,601, a conservative p-value of 0.75, and a 90/10 confidence/precision level. This yielded a minimum requirement of 91 households. The Project Proponent conducted 1,188 surveys, exceeding the requirement by more than 13 times, thereby ensuring high statistical robustness. The VVB reviewed the sample size calculation /10/ and survey implementation records /08/ and confirmed their alignment with the approved methodology and monitoring plan.

$$n \geq \frac{1.645^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.645^2 \times p(1-p)} \quad \text{Equation 5}$$

The survey results produced a usage rate of 1.0. However, as per the *Corrections and Clarifications to VM0050* (February 2025), this parameter must be capped at 0.9 if three specific conditions are met. The VVB verified that:

- **Technology selection:** Local cooking practice and fuel-use assessments confirmed that the ICS model fully meets household cooking needs, including for meal preparation and water boiling.
- **User support:** Dedicated repair teams and a local field office were established in Instance 2, with user training and instructional materials provided in local languages.
- **Accessibility:** A toll-free helpline and grievance mechanisms are operational and accessible, ensuring households can request support and maintenance as needed.

The VVB reviewed the sample size calculation, household survey records, survey implementation protocols, evidence of technology suitability for local cooking practices, user support mechanisms (training and instructional materials in local languages), maintenance and repair systems (including field office and spare part management), accessibility mechanisms (toll-free helpline and grievance register), and compliance with the *Corrections and Clarifications to VM0050* (Feb 2025). Based on this review, the VVB concludes that the capped value of $n_{j,k,y}$ = 0.9 is appropriate, justified, and in full compliance with *Correction and Clarifications to VM0050 Energy Efficiency and Fuel Switch Measures in Cookstoves, v1.0 /B07/*.

BC_{b,i,y}, and BC_{p,j,k,y}

Parameters BC_{b,i,y}, and BC_{p,j,k,y} were monitored using the Kitchen Performance Test (KPT) Protocol v4.0, in accordance with VM0050 v1.0. A paired-sample study design was applied, wherein the same households were surveyed in both the baseline and project scenarios, enabling direct comparison of fuel use.

As per VM0050 requirements, the KPT results must achieve a minimum 90% confidence level with 10% precision. Where this threshold is not initially met, additional KPTs are required until the criterion is satisfied. The VVB reviewed the monitoring approach, raw data, statistical analyses, and supporting spreadsheets (“ICS-01 KPT Instance 1” and “ICS-01 KPT Instance 2”) /22/ and confirms that the applied approach is methodologically sound and aligned with the approved methodology and monitoring plan.

During the current monitoring period, KPTs were conducted in Instance 2, following the completion of activities in Instance 1. The minimum required sample size was calculated using the equation in *Guideline: Sampling and surveys for CDM project activities and PoAs, Appendix 1, paragraph 12*:

$$n \geq \frac{1.645^2 N \times V}{(N - 1) \times 0.1^2 + 1.645^2 \times V}$$

$$V = \left(\frac{SD}{mean} \right)^2$$

Equation 6

Assuming a population size of **115,601**, a conservative **p-value of 0.75**, and the 90/10 confidence/precision requirement, the minimum required sample size was **36 households**. In practice, the Project Proponent conducted **150 project KPTs in instance 1 and 144 project KPTs in instance 2** and **151 baseline KPTs in instance 1 and 154 baseline KPTs in instance 2** households far exceeding the minimum requirement. The VVB reviewed the sample size calculation /10/ and survey implementation records /08/, confirming that the sample design, implementation, and documentation comply with approved protocols.

Results verified as follows:

- **Instance 1:** BC_{b,i,y} = 4.83 t/year; BC_{p,j,k,y} = 4.55 kg/day (1.66 t/year).
- **Instance 2:** BC_{b,i,y} = 6.04 t/year; BC_{p,j,k,y} = 4.98 kg/day (1.82 t/year).

The statistical analysis for both baseline and project KPTs was replicated and validated. For baseline KPTs (BC_{b,i,y}) in Instance 2, 154 samples yielded a mean of 5.50 kg/pp/day and a standard deviation of 1.78 kg/pp/day, equivalent to 6.04 tonnes/year, resulting in a precision of 4.28%, which is well within the 10% threshold and meets the 90/10 confidence/precision requirement.

For project KPTs (BC_{p,j,k,y}), Instance 1 (150 samples) had a mean of 1.34 kg/pp/day and standard deviation of 0.34 kg/pp/day (1.66 tonnes/year) with a precision of 3.41%, and Instance 2 (144 samples) had a mean of 1.64 kg/pp/day and standard deviation of 0.45 kg/pp/day (1.82 tonnes/year) with a precision of 3.72%; both results satisfy the methodology criteria.

The VVB concludes that the monitoring and statistical analyses for BC_{b,i,y} and BC_{p,j,k,y} were conducted in accordance with the approved methodology VM0050 v1.0 /B02/, the KPT Protocol v4.0 /B08/, and the project monitoring plan. The results are statistically robust, exceed minimum sampling requirements, and meet the 90/10 confidence/precision criteria. All calculated precisions are below the 10% threshold, confirming that the KPT results are reliable, appropriate, and suitable for verification and emission reduction estimation.

Hh_i and Hh_{j,k}

The verification team assessed the monitoring of parameters **Hh_i** (average baseline household size in male adult equivalent units) and **Hh_{j,k}** (average project household size). Household composition was recorded during the Kitchen Performance Test (KPT) surveys, converted into equivalent male adult units in line with the **Kitchen Performance Test Protocol v4.0**, and averaged across surveyed households.

In accordance with **VM0050 v1.0/B02/** and the CDM sampling guidelines, these parameters are required to achieve a minimum confidence level of 90% with 10% precision. The verification team reviewed the household survey forms, conversion records, statistical calculations, and monitoring database.

For **baseline households (Hh_i)**, 154 samples were analyzed, resulting in a mean of 3.13 and a standard deviation of 1.14. The calculated precision was **4.83%**, meeting the 90/10 requirement. For **project households (Hh_{j,k})**, 144 samples were analyzed, resulting in a mean of 1.64 and a standard deviation of 0.45. The calculated precision was **3.72%**, also meeting the 90/10 requirement.

The verification team concludes that the monitoring and calculation of **Hh_i** and **Hh_{j,k}** were conducted in compliance with the requirements of the approved methodology and monitoring plan. The parameters are statistically robust and acceptable for use in the emission reduction calculations.

The parameter **η_{new,avg,y}**, representing the average thermal efficiency of stoves used for at least one year (y=1), was monitored in accordance with the approved **VM0050 v1.0 methodology /B02/**. Efficiency tests were conducted by the independent laboratory **Koshish Sustainable Solutions Pvt Ltd** in September 2024. Six ICS were selected using a Simple Random Sampling approach as described in section 5.3.2 of the PD, and each ICS was tested three times, yielding a total of 18 WBT results. The measured efficiencies ranged from 34.5% to 35.0%, with a calculated average of 34.73%. All results, including QR codes, are documented in Appendix 2 of MR. Statistical analysis using the Student's t-distribution for small sample sizes confirmed the 90% confidence level and precision of the measurements, with a margin of error of 0.094, a confidence interval of 34.73 ± 0.094, and a calculated precision of 0.27%, well below the 10% threshold required by the methodology.

Based on the review of the sampling approach, test execution, and statistical analysis, the VVB concludes that the procedures applied for monitoring **η_{new,avg,y}** are consistent with the approved methodology and CDM guidelines. The measured average efficiency of 34.73% is considered reliable, and the mandatory 90/10 confidence/precision requirements are fully satisfied. The verified parameter can therefore be used for the calculation of emission reductions.

also reviewed the video evidence and software screenshots demonstrating the random sampling process for ICS selection /11/. The sample size calculations, monitoring methodologies, and field data collection are confirmed to be in full compliance with the registered monitoring plan and VM0050 methodology requirements.

The sample size calculation is summarised in the table below :

Monitored Parameter	Sample size	Actual Samples	Precision Achieved
$n_{j,k,y}$	91	1188	90/10
$BC_{b,i,y}$	36	305	90/10
$BC_{p,j,k,y}$	36	294	90/10
$Hh_i / Hh_{j,k}$	36	599	90/10
$\eta_{new,avg,y}$	3	6	90/10

The verification team concludes that the PP has implemented the sampling and surveys in accordance with the registered sampling plan, and the approach adopted for sample size estimation and field implementation is compliant with all applicable requirements. All underlying data and survey records have been cross-verified and are found to be accurate, consistent, and acceptable for verification purposes.

During verification, VVB has applied acceptance sampling to determine the operational status of the ICS in the households and adjustment to account for any continued use of pre- project devices during the year y. A sample size of 11 cookstove from both instance was chosen using paragraph 39 (c) of the sampling standard, version 09 /B04/. A sample size of 11 was determined, based on an AQL of 0.5% and UQL of 20%, producer risk 10% and consumer risk 10%. Acceptance number (c) thus determined for the sample is 0. Accordingly, VVB interviewed 22 Households (11 from each instance). It was observed that out of the 22 stoves inspected, all the 22 stoves were found to be operational and this matched with the PP's records and hence no discrepant records were observed with the MR /01/ and ER sheet /02/ and thus c=0. Thus, PP's set of records has been accepted in line with § 33 of the sampling standard, version 09.0/B04/. Verification team has cross verified these sample documents.

In line with Paragraph 25 of the CDM Sampling Standard, version 09.0 /B04/, the verification team has assessed the implementation of the sampling plan against the criteria specified in the registered monitoring plan. This included an evaluation of:

(a) Whether the required confidence/precision level has been met – confirmed through review of the sampling calculation and survey data;

(b) Whether the selected sample is representative of the population – confirmed through documentation checks and review of the applied simple random sampling methodology.

Based on this verification, the sampling activities and associated calculations carried out by the PP are found to be consistent with the applied methodology, monitoring plan, and relevant sampling guidelines. All records were reviewed and found to be traceable and complete.

Table 2:- Parameter selected during Monitoring

Monitored Parameter	How the PP conducted sampling surveys (to obtain the project participants' or the coordinating/managing entities' records)	How the VVB could obtain records for verification	Criteria for deciding what ultimately constitutes a discrepancy
$n_{j,k,y}$	Sampling based survey (questionnaire survey/interviews) Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use [Yes/No]	Cross-check of a sample of project participants' samples (questionnaire operation surveys/interviews) including but not limited to following: Consistency between the information as contained in Survey sheet and revealed from the on-site interviews. Enquire/observe the project stove and its operation during the project scenario.	VVB results, accounting for duly justified differences.
$BC_{b,i,y}$	Sampling-based survey (structured monitoring surveys) KPT conducted on the selected households sampled from project households on the baseline stove to calculate average fuel consumption for traditional stoves	A representative sample of project participants' Kitchen Performance Test (KPT) records will be cross-checked to assess data accuracy and determine whether the 90/10 confidence/precision requirement is met. Verification activities will include:	VVB results, accounting for duly justified differences.

	Visual inspection of kitchen conditions and practices; Test results must meet 90/10 confidence/precision level; additional samples collected if required [Yes/No]	- Comparing recorded KPT data with user recall during on-site interviews. - Observing traditional stove use and kitchen conditions to validate baseline fuel consumption data	
$BC_{p,j,k,y}$	Sampling-based survey (in-person KPTs conducted in sample households) KPT conducted on the selected households sampled from project households on the baseline stove to calculate average fuel consumption for traditional stoves Visual inspection of kitchen conditions and technologies; interviews with end users to verify consistency with baseline behavior [Yes/No] Test results must meet 90/10 confidence/precision level; additional samples collected if required [Yes/No]	A sample of project participants' monitored data—including KPT results, interviews, and field observations—will be cross-checked to ensure accuracy and compliance, using a sampling approach designed to meet a 90/10 confidence/precision level. This will include: - Assessing consistency between monitored KPT data, stove usage logs, and information obtained during user interviews. - Observing improved cookstove (ICS) usage and verifying alignment with the approved monitoring plan and VM0050 v1.0 methodology to confirm project fuel consumption.	VVB results, accounting for duly justified differences.
$Hh_i / Hh_{j,k}$	Sampling-based survey (Kitchen Performance Test - KPT) conducted by trained field staff Household composition recorded during surveys to calculate equivalent standard male adult per household, following FAO Woodfuel Survey Guidelines	Cross-check of household survey data and sampling protocols, including but not limited to the following: - Consistency between reported household size data and regional census benchmarks and sample households (household size) - Review of sampling methods and calculations to ensure	VVB results, accounting for duly justified differences.

	Average household size derived from the sample data [Yes/No]	alignment with VM0050 v1.0 methodological requirements	
$\eta_{new,avg,y}$	Sampling-based testing conducted by an independent technical laboratory For each ICS type i and vintage y, 3 ICS units were sampled from the first batch and subjected to 3 WBTs each (total of 9 tests) Certified WBT results obtained from experienced technical laboratories Degradation of efficiency from the first batch applied to all batches [Yes/No] Test results required to meet methodology criteria for standard deviation and 90/10 confidence/precision levels; additional testing conducted if criteria not met [Yes/No]	Review of efficiency test documentation and supporting data, including but not limited to the following: • Verification of WBT reports from independent laboratories and manufacturer specifications • Confirmation that the weighted average thermal efficiency meets performance criteria under VM0050 v1.0 and sampling required to meet the standard deviations limit of 0.05 and fit the 90/10 confidence/precision level	VVB results, accounting for duly justified differences.

The sampling plan implemented by the PP is in accordance with the applied approved monitoring methodology /B02/ and the VCS PD /03/. The PP has appropriately performed Simple random Sampling procedure in line with the applied methodology. As the VCS PD/03/ mentions the option for Simple random sampling procedure, it is acceptable to the verification team.

The necessary confidence / precision of 90/10 each of the parameters are met. This has been cross verified by the verification team from the supporting documents submitted/10/

Emission reductions have been calculated in accordance with the applied methodology VM0050 version 1.0 /B02/, and VCS PD /03/. The PP has used monitored data and ex-ante fixed data including default values as mandated/permitted by the applied methodology. The values used for calculation of GHG emission reductions have been thoroughly checked by the verification team and was found appropriate and correct.

Table 3:- Parameters Determined ex-ante:

The following parameters are determined ex-ante and mentioned in section 5.1 of the VCS PD/03/,

Parameter	Unit	Value	Assessment
$EF_{b,i,CO2}$	tCO ₂ e/TJ	112	– Fixed ex-ante – Default values from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion have been used.
$EF_{p,j,CO2}$	tCO ₂ e/TJ	112	– Fixed ex-ante – Default values from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion have been used.
$EF_{b,i,nonCO2}$	tCO ₂ e/TJ	9.46	– Fixed ex-ante – Default values from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion have been used.
$EF_{p,j,nonCO2}$	tCO ₂ e/TJ	9.46	– Fixed ex-ante – Default values from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion have been used.
$NCV_{b,i}$	TJ.tonne	0.0156	– Fixed ex-ante – Default values from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction have been used.
$NCV_{p,j}$	TJ.tonne	0.0156	– Fixed ex-ante – Default values from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction have been used.
$BC_{ex-ante,b,i}$	Tonnes	4.83	– Fixed ex-ante – The value is based on average quantity of fuel used per baseline device type i per

			average number of adults in household per year
$\eta_{old,avg}$	Fraction	15%	– Fixed ex-ante – The default value for three-stone fires using firewood or cookstoves,,,,,,

The spread sheet submitted by the PP clearly and transparently mentions values of the data parameters used for calculation of emission reductions. The input values have been verified from the reliable and authentic sources including monitoring records (distribution records) /09/, Monitoring Report /01/, and applied methodology /B02-a/. The emission reductions calculated were compared with the emission reduction spread sheet /02/ and found to be correct. No significant reporting risks have been identified for the data reported.

The details of monitoring parameters used for calculation of emission reductions are provided below.

Table 4:- Parameters monitored ex-post

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in monitoring plan of VCS PD):	$N_{j,k,y}$ Number of commissioned project devices of type j from batch k in year y
Measuring frequency/Time Interval:	At the time of commissioning/distribution of ICS
Reporting frequency:	At least once every two years
Reported value:	Instance 1: 62,121 ICS Instance 2: 53,480 ICS Total : 115,601
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Details of monitoring equipment:	Value obtained from monitoring survey of samples /08/.
Is accuracy of the monitoring equipment as stated in the VCS PD? If the VCS PD does not specify the accuracy of the monitoring	NA

equipment, does the monitoring equipment represent good monitoring practise?	
Calibration frequency /interval: Is it monitoring methodology /CDM EB guidance / local or national standards / manufacturers specification	NA
Is the calibration interval in line with the monitoring plan of the VCS PD? If the VCS PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	NA. QA/QC procedures stated in MR comply with VCS PD /03/
Company performing the calibration(internal or external calibration):	NA
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	NA
Is (are) calibration(s) valid for the whole reporting period?	NA
If applicable, has the reported data been cross-checked with other available data?	Yes, the reported data in MR has been compared with monitoring survey records /08/ and the ER sheet /02/.
How were the values in the monitoring report verified?	NA
Does the data management (from data generation to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data management ensures correct transfer of data and reporting of emission reductions and all necessary QA/QC processes are in place.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	NA

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in monitoring plan of VCS PD):	$n_{j,k,y}$ Proportion of commissioned project devices of type j from batch k that are still being used regularly in year y
Measuring frequency/Time Interval:	At least annually
Reporting frequency:	At least annually
Reported value:	0.9
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Details of monitoring equipment:	Value obtained from monitoring survey of samples using tablets with the Tramontana ICS software application /08/.
Is accuracy of the monitoring equipment as stated in the VCS PD? If the VCS PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practise?	NA
Calibration frequency /interval: Is it monitoring methodology /CDM EB guidance / local or national standards / manufacturers specification	NA
Is the calibration interval in line with the monitoring plan of the VCS PD? If the VCS PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	NA. QA/QC procedures stated in MR comply with VCS PD /03/
Company performing the calibration(internal or external calibration):	NA

Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	NA
Is (are) calibration(s) valid for the whole reporting period?	NA
If applicable, has the reported data been cross-checked with other available data?	Yes, the reported data in MR has been compared with monitoring survey records /08/ and the ER sheet /02/.
How were the values in the monitoring report verified?	NA
Does the data management (from data generation to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data management ensures correct transfer of data and reporting of emission reductions and all necessary QA/QC processes are in place.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	NA

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in monitoring plan of VCS PD):	(Hh_i) Average household size of the target population using baseline device type i $(Hh_{j,k})$ Average household size of the target population using project device type j from batch k
Measuring frequency/Time Interval:	A new value is established each time the project activity is implemented in a new instance.
Reporting frequency:	At least annually

Reported value:	For the first project instance: $Hh_i = 3.38 \text{ (includes control households)}$ $Hh_{j,k} = 3.47$ For the second project instance: $Hh_i = 3.13 \text{ (includes control households)}$ $Hh_{j,k} = 3.09$
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Details of monitoring equipment:	Value obtained from monitoring survey of samples using tablets with the Tramontana ICS software application /08/.
Is accuracy of the monitoring equipment as stated in the VCS PD? If the VCS PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practise?	NA
Calibration frequency /interval: Is it monitoring methodology /CDM EB guidance / local or national standards / manufacturers specification	NA
Is the calibration interval in line with the monitoring plan of the VCS PD? If the VCS PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	NA. QA/QC procedures stated in MR comply with VCS PD /03/
Company performing the calibration (internal or external calibration):	NA
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	NA
Is (are) calibration(s) valid for the whole reporting period?	NA
If applicable, has the reported data been cross-checked with other available data?	Yes, the reported data in MR has been compared with monitoring survey records /08/ and the ER sheet /02/.

How were the values in the monitoring report verified?	NA
Does the data management (from data generation to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data management ensures correct transfer of data and reporting of emission reductions and all necessary QA/QC processes are in place.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	NA

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in monitoring plan of VCS PD):	(n_{HH}), Number of ICS distributed to each household
Measuring frequency/Time Interval:	At the time of commissioning/distribution of ICS
Reporting frequency:	for every new instance
Reported value:	1
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Details of monitoring equipment:	Value obtained from monitoring survey of samples using tablets with the Tramontana ICS software application /08/.
Is accuracy of the monitoring equipment as stated in the VCS PD? If the VCS PD does not specify the accuracy of the monitoring	NA

equipment, does the monitoring equipment represent good monitoring practise?	
Calibration frequency /interval: Is it monitoring methodology /CDM EB guidance / local or national standards / manufacturers specification	NA
Is the calibration interval in line with the monitoring plan of the VCS PD? If the VCS PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	NA. QA/QC procedures stated in MR comply with VCS PD /03/
Company performing the calibration(internal or external calibration):	NA
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	NA
Is (are) calibration(s) valid for the whole reporting period?	NA
If applicable, has the reported data been cross-checked with other available data?	Yes, the reported data in MR has been compared with monitoring survey records /08/ and the ER sheet /02/.
How were the values in the monitoring report verified?	NA
Does the data management (from data generation to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data management ensures correct transfer of data and reporting of emission reductions and all necessary QA/QC processes are in place.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	NA

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in monitoring plan of VCS PD):	$(f_{NRB,y})$, fraction of woody biomass that is established to be non-renewable used by device in year y
Measuring frequency/Time Interval:	A new value is established each time the project activity is implemented in a new instance.
Reporting frequency:	At the time of commissioning/distribution of ICS
Reported value:	– Instance 1: A value of 0.510 was calculated. – Instance 2: A value of 0.556 was calculated.
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Details of monitoring equipment:	Value obtained from baseline survey of project activity that is implemented in a new instance and calculations/08/.
Is accuracy of the monitoring equipment as stated in the VCS PD? If the VCS PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practise?	NA
Calibration frequency /interval: Is it monitoring methodology /CDM EB guidance / local or national standards / manufacturers specification	NA
Is the calibration interval in line with the monitoring plan of the VCS PD? If the VCS PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	NA. QA/QC procedures stated in MR /01/ comply with VCS PD /03/

Company performing the calibration(internal or external calibration):	NA
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	NA
Is (are) calibration(s) valid for the whole reporting period?	NA
If applicable, has the reported data been cross-checked with other available data?	Yes, the reported data in MR has been compared with monitoring survey records /08/ and the ER sheet /02/.
How were the values in the monitoring report verified?	NA
Does the data management (from data generation to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data management ensures correct transfer of data and reporting of emission reductions and all necessary QA/QC processes are in place.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	NA

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in monitoring plan of VCS PD):	$BC_{b,i,y}$ Average quantity of fuel used per baseline device type i during year y
Measuring frequency/Time Interval:	A new value is established each time the project activity is implemented in a new instance.
Reporting frequency:	Every five years

Reported value:	– Instance 1: An average fuel consumption of 13.22 kg per day per baseline stove was measured, equivalent to 4.83 tonnes annually. – Instance 2: An average fuel consumption of 16.55 kg per day per baseline stove was measured, equivalent to 6.04 tonnes annually.
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Details of monitoring equipment:	Value obtained from samples from KPT survey of both instances /22/.
Is accuracy of the monitoring equipment as stated in the VCS PD? If the VCS PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practise?	NA
Calibration frequency /interval: Is it monitoring methodology /CDM EB guidance / local or national standards / manufacturers specification	NA
Is the calibration interval in line with the monitoring plan of the VCS PD? If the VCS PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	NA. QA/QC procedures stated in MR comply with VCS PD /03/
Company performing the calibration(internal or external calibration):	Yes, The digital moisture meters and weighing scales have been calibrated by the manufacturer as per monitoring equipment manual /27/
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	Yes
Is (are) calibration(s) valid for the whole reporting period?	NA

If applicable, has the reported data been cross-checked with other available data?	Yes, the reported data in MR has been compared with monitoring survey records /08/ and the ER sheet /02/.
How were the values in the monitoring report verified?	NA
Does the data management (from data generation to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data management ensures correct transfer of data and reporting of emission reductions and all necessary QA/QC processes are in place.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	NA

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in monitoring plan of VCS PD):	$\eta_{new,avg,y}$ Weighted average efficiency of project devices in year y
Measuring frequency/Time Interval:	Annually
Reporting frequency:	Annually
Reported value:	– For y=1, the thermal efficiency of the CRS-23 project measured from tests conducted on 6 randomly selected operational project ICS resulted in an average value of 34.73%
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Details of monitoring equipment:	Value obtained from WBT test report provided by independent agency /21/.

Is accuracy of the monitoring equipment as stated in the VCS PD? If the VCS PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practise?	NA
Calibration frequency /interval: Is it monitoring methodology /CDM EB guidance / local or national standards / manufacturers specification	NA
Is the calibration interval in line with the monitoring plan of the VCS PD? If the VCS PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	QA/QC procedures stated in MR comply with VCS PD /03/.
Company performing the calibration(internal or external calibration):	NA
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	NA
Is (are) calibration(s) valid for the whole reporting period?	NA
If applicable, has the reported data been cross-checked with other available data?	Yes, the reported data in MR has been compared with monitoring survey records /08/ and the ER sheet /02/.
How were the values in the monitoring report verified?	NA
Does the data management (from data generation to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data management ensures correct transfer of data and reporting of emission reductions and all necessary QA/QC processes are in place.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption	NA

theoretically possible been applied or has a request for deviation been approved?	
---	--

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in monitoring plan of VCS PD):	$BC_{p,j,k,y}$ Average quantity of fuel used by project device type j from batch k during year y
Measuring frequency/Time Interval:	Annually
Reporting frequency:	Annually
Reported value:	– Instance 1: an average value of 4.55 kg per day was obtained for the project ICS, which is equivalent to 1.66 tonnes annually. – Instance 2: an average value of 4.98 kg per day was obtained for the project ICS, which is equivalent to 1.82 tonnes annually
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Details of monitoring equipment:	Value obtained from monitoring survey of samples /08/.
Is accuracy of the monitoring equipment as stated in the VCS PD? If the VCS PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practise?	NA
Calibration frequency /interval: Is it monitoring methodology /CDM EB guidance / local or national standards / manufacturers specification	NA

Is the calibration interval in line with the monitoring plan of the VCS PD? If the VCS PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	NA. QA/QC procedures stated in MR comply with VCS PD /03/
Company performing the calibration(internal or external calibration):	Yes, The digital moisture meters and weighing scales have been calibrated by the manufacturer as per monitoring equipment manual /27/
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	Yes
Is (are) calibration(s) valid for the whole reporting period?	NA
If applicable, has the reported data been cross-checked with other available data?	Yes, the reported data in MR has been compared with monitoring survey records /08/ and the ER sheet /02/.
How were the values in the monitoring report verified?	NA
Does the data management (from data generation to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data management ensures correct transfer of data and reporting of emission reductions and all necessary QA/QC processes are in place.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	NA

Verification team confirms that all parameters are used correctly in the calculations, all results are verifiable and transparent, all assumptions are described and based on verifiable evidence and calculations are done in accordance with the pre-defined formulae from registered VCS PD /03/. The total number of emission reductions for the monitoring period (24 – August - 2023 to 12 – September -2024) is 172,045 tCO₂e.

Table 6: Emission reduction claimed during this monitoring period

Vintage period	Baseline em	Project emissions (t	Leakage emissi	Reduction VCU	Total VCUs (tCO ₂ e)
21-Jan-2023 to 31-Dec-2023	118,907	49,531	3,469	65,907	65,907
01-Jan-2024 to 12-Sep-2024	245,643	103,253	7,120	135,271	135,271
Total	364,550	152,783	10,588	201,178	201,178

The verification team reviewed and confirmed the accuracy of the emission reduction calculations provided in the spreadsheet. The monitoring report is supported by this spreadsheet, and all formulas and data consistency were verified and found to be correct. A comparison of ex-ante and ex-post emission reductions, as presented in Section 5.4 of the Monitoring Report (MR/01/), clearly indicates that the achieved reductions are lower than originally estimated. This is primarily due to the fact that not all cookstoves were operational from the start of the crediting period. This explanation was validated during the onsite verification, and the remark provided by the project proponent is deemed appropriate and justified.

4.4 Quality of Evidence to Determine Reductions and Removals

Carbon Check verified the reported emission reductions by ensuring a clear and traceable audit trail supported by appropriate evidence and records. All source documents underpinning the GHG data, including assumptions and calculations, were reviewed.

In assessing the audit trail, Carbon Check examined:

1. The availability and adequacy of evidence across the full monitoring period
2. The origin and reliability of the source data
3. Cross-verification of figures against independent or alternative sources, as detailed in Appendix 1

The data collection system was found to be in line with the monitoring plan and the applied methodology. Proper data acquisition, aggregation, and management protocols were in place. Monitoring personnel were well-trained, followed standardized procedures, and demonstrated sufficient competence to ensure accurate data collection and reporting.

4.5 Non-Permanence Risk Analysis

The project activity remained operational throughout the entire monitoring period. As a result, a non-performance analysis rating was not required for this monitoring period.

5 VERIFICATION OPINION

5.1 Verification Summary

The Project Proponent, Tramontana Impact Limited, commissioned Carbon Check (India) Private Limited (Carbon Check) to perform the 2nd periodic verification of the VCS Project Activity titled “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” (VCS ID – 4457) for the monitoring period from 24 - August - 2023 to 12 - September - 2024. Tramontana Impact Limited holds the responsibility for data collection in line with the monitoring plan and for documenting the greenhouse gas (GHG) emission reductions resulting from the project activity.

This report summarizes the findings of the verification conducted in accordance with the latest VCS criteria (VCS Standard Version 4.7) /B01/, and relevant requirements for consistent project implementation, monitoring, and reporting.

Based on the documentation reviewed and the findings of the on-site assessment, Carbon Check confirms that:

- The project activity has been implemented with high quality and operated in accordance with the registered design and the approved monitoring plan, while keeping all process clear and easy to track.
- The project activity strictly follows VCS rules and requirements and fully conforms to all provisions set out in the latest version of the applicable methodologies.
- The monitoring report /01/ and supporting documentation are complete, transparent, and in accordance with the applicable VCS and methodological requirements.
- The monitoring plan features a solid framework that enables thorough oversight of the project’s progress.
- The monitoring system is in place and consistent with the approved baseline and monitoring methodology.
- The project ensures the uniqueness of all ICS distributed through a Tramontana ICS software and user photos. The assessment team confirmed no duplicate or repeated households. The VVB’s extensive site visit verified household coordinates via geographic mapping and cross-checked government ID with the distribution database, confirming data accuracy.

In Carbon Check’s opinion, the GHG emission reductions stated in the monitoring report version 1.3 dated 02-January-2026 for the monitoring period from 24-August-2023 to 12-September - 2024 have been calculated accurately and fairly presented. The emission reductions are determined in accordance with the applied methodology and the VCS Standard Version 4.7.

The verification team concludes that the project activity, as described in the VCS PD and the Monitoring Report, meets all applicable VCS requirements and that the verification was conducted in accordance with VCS rules and ISO 14064-3:2019.

5.2 Verification Conclusion

Carbon Check concludes the verification with a positive opinion that the VCS Project Activity titled “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India,” as presented in the monitoring report version 1.3 dated 02-January-2026 /01/, complies with all applicable requirements of the Verified Carbon Standard (VCS), including those outlined in the VCS Standard, the applied methodology, and relevant tools and guidelines.

The applied baseline and monitoring methodology /B02/ (VM0050, Version 1.0) is appropriate for the project and has been correctly implemented. The verification team confirms that the project activity has been executed and monitored in accordance with the approved Monitoring Report /01/ and the monitoring plan.

Verification Period: 24-August-2023 to 12-September-2024

Verified GHG Emission Reductions: 172,045 tCO₂e for the stated verification period.

Verified GHG emission reductions and carbon dioxide removals in the above verification period

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCU (tCO ₂ e)	Total VCUs (tCO ₂ e)
24-Aug-2023 to 31-Dec-2023	66,347	27,637	1,935	36,775	36,775
01-Jan-2024 to 12-Sep-2024	245,643	103,253	7,120	135,270	135,270
Total	311,990	130,890	9,055	172,045	172,045

Based on the assessment, the VVB is of the opinion that the project has been implemented in accordance with the registered/revised PD. The Monitoring Plan is consistent with the approved methodology, and the monitoring activities have been conducted in compliance with the MP. The monitored data and the calculation of emission reductions (ERs) have been assessed and confirmed to be accurate and in accordance with VCS requirements.

Accordingly, Carbon Check (India) Private Limited certifies the verified emission reductions for the monitoring period from 24-August-2023 to 12-September-2024, amounting to 172,045 tCO₂e, and recommends the issuance of the corresponding Verified Carbon Units (VCUs) by the VCS Registry.

5.3 Ex-ante vs Ex-post ERR Comparison

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
24-August-2023 to 31-December-2023	67,034	36,775	−45.1%	Due to the conservative cap applied to baseline energy consumption ($EC_{(i,y)}$) under VM0050. While 0.0753 TJ was used ex-ante based on stove efficiencies, a capped value of 0.0621TJ was applied ex-post using KPT results, resulting in lower emission reductions.
01-January-2024 to 12-September-2024	283,923	135,271	−52.4%	During the 2024 monitoring period, the difference in emission reductions was primarily due to extreme weather from June to September, which limited access to the project area and reduced daily stove distribution from approximately 900 to 5 units per day. Distribution was also temporarily suspended during India's general elections between April and June 2024. Additional factors include the exclusive distribution of CRS-23 stoves (whereas the ex-ante model assumed STAR-25 deployment) and improved survey implementation, which reduced data collection errors. The VVB considers these factors reasonable and adequately explains the deviation from ex-ante projections.
Total	350,957	172,045	−51.0%%	

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No Commercially sensitive information provided by the PP during this monitoring period.

APPENDIX 2: SUPPORTING DOCUMENTS

APPENDIX 2.1: REFERENCE DOCUMENTS

Ref	Document
/01/	VCS monitoring report version 1.3 dated 02-January-2026
/02/	ER calculation spread sheet
/03/	VCS PD for the project “TRAMONTANA ICS-01 – The distribution of improved cookstoves to indigenous communities across rural India” version 1.8, dated 02-January-2026.
/04/	Registration cum consent deed as evidence for the start date of the project.
/05/	Registration cum consent deed as evidence for unique identification of each of the ICS.
/06/	Proof of right of VERs.
/07/	Monitoring survey questionnaire template.
/08/	Monitoring survey records for the monitoring period. ICS-01 Monitoring Database M2
/09/	Database for the ICS distribution records for the monitoring period.
/10/	Sample size calculation and precision level achieved calculator for the monitoring period.
/11/	Video recording of the random sample generator software as evidence for random sample selection for the monitoring survey parameters.
/12/	Training records; 04-July-2024 - Attendance register - Training Images
/13/	Scanned grievance logbook/register for MP2 (24-August-2023 to 12-September-2024)
/14/	Company registration certificate ‘Sequoia Plantation Togo SAU’ .
/15/	PPs declaration that; - Project is not part of any other GHG program. - Project will not claim emission reductions under any other program. - No project device will be double counted. - PP has right of verified emission reductions.
/16/	Technical specifications of the ICS including the life span. CRS-23 model ICS

	STAR-25 model ICS
/17/	Evidence of employment generated during MP2.
/18/	Manufacturer's brochure of ICS
/19/	Onsite visit documents
/20/	Evidence for fNRB calculation - fNRB calculation_instance 1 - fNRB calculation_instance 2
/21/	Water Boiling Tests (WBT) Reports conducted by " <i>Thermodynamics Laboratory at the Department of Mechanical Engineering, Jorhat Engineering College (JEC)</i> " dated 17-January-2020.
/22/	Kitchen performance test (KPT) database - ICS-01 KPT Instance 1 and - ICS-01 KPT Instance 2
/23/	Baseline database - ICS-01 Baseline Database Instance 1 - ICS-01 Baseline Database Instance 2
/24/	Third Party Testing Certificate for KPT conducted by Koshish Sustainable Solutions Pvt. Ltd.
/25/	Stakeholder consultation minutes, attendances for instance 2, Stakeholder input records
/26/	CropZone Human Resource Policy Manual/ HR policy
/27/	Calibration of KPT tools - Digital Scales, Atom Aliston - Moisture meter
/28/	KML file for project boundary

APPENDIX 2.2: BACKGROUND REFERENCE DOCUMENTS

Ref	Document
/B01/	VCS Requirements a. VCS Standard (v4.7, dated 16-April-2024) b. VCS Program Guide (v4.4, dated 29-August-2023) c. VCS Validation and Verification Manual version (v3.2, dated 19-October-2016) d. Registration & Issuance Process (v4.5, dated 16-April-2024) e. VCS Program Definitions version (v4.5, dated 16-April-2024) f. VCS MR template version 4.4
/B02/	Applied baseline and monitoring methodology. a. VM0050. version 1.0, “Energy Efficiency and Fuel-Switch Measures In Cookstoves” b. CDM AMS-II G Version 13.1 “Energy efficiency measures in thermal applications of non-renewable biomass”
/B03/	Methodological Tool • VT0008 Version 1.0 “Additionality Assessment” dated 14-October-2024 • CDM Tool 30 Version 04.0 “Calculation of the fraction of non-renewable biomass”
/B04/	a. “Standard for sampling and surveys for CDM project activities and programme of activities” (version 09.0) b. Guidelines for sampling and surveys for CDM project activities and Programme of Activities (version 04)
/B05/	Website and links: 1. IPCC (http://www.ipcc-nggip.iges.or.jp) 2. http://cdm.unfccc.int 3. http://www.v-c-s.org
/B06/	Previous Methodology • VMR0006. version 1.1, “Methodology for Installation of High Efficiency Firewood Cookstoves”
/B07/	Correction And Clarifications To VM0050 Energy Efficiency and Fuel Switch Measures In Cookstoves, V1.0 Publication date: 03 February 2025

/B08/	Kitchen Performance Test Protocol Version 4.0
/B09/	India's Child and Adolescent Labour Act,1986

APPENDIX 3: ABBREVIATIONS

ASP	Acceptance Sampling Plan
BE	Baseline Emission
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification Request
CO₂	Carbon Dioxide
CO_{2e}	Carbon Dioxide Equivalent
DVR	Draft Verification Report
EF	Emission Factor
ER	Emission Reduction
FAR	Forward Action Request
FVR	Final verification Report
GHG	Greenhouse gas(es)
ICS	Improved Cookstoves
IPCC	Intergovernmental Panel on Climate Change
KPT	Kitchen Performance Test
NA	Not Applicable
OSV	On Site Visit
PD	Project Description
PMUY	Pradhan Mantri Ujjwala Yojana
PP	Project Proponent
QA/QC	Quality assurance/ Quality control
SDG	Sustainable Development Goals
TR	Technical Review
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit
VVB	Validation Verification Body
VVM	Validation and Verification Manual
WBT	Water Boiling Test

APPENDIX 4: CERTIFICATES OF COMPETENCE

Carbon Check (India) Private Limited

Certificate of Competency

Manas Halder

has been qualified as per CCIPL's internal qualification procedures in accordance with the requirements of CDM AS, A 6.4 AS/ ISO/IEC14065:2020, ISO/IEC 17029:2019 and other applicable GHG programs:

for the following functions and requirements:

- | | | | |
|--|---|--|---|
| ☐ Management Function | ☐ Administrative | ☒ Validator/Verifier | ☒ Team Leader |
| ☐ Acting Team Leader | ☒ Technical Expert | ☐ Technical Reviewer | ☐ Trainee Assessor |
| ☒ Assessor | ☒ Regional Expert | ☐ Gender Expert | ☐ Financial Expert |
| | Bangladesh, I | | |
| ☐ Plastic Waste Expert | ☐ Health Expert | ☐ CCB Expert | ☒ SDG Expert |
| ☒ Expert Social Aspect | ☒ Expert Environmental Aspect | ☐ Internal Auditor | ☐ Legal Expert |
| ☒ FOEN Approved Technical Expert | ☐ FOEN Approved Quality Officer | | |

in the following Technical Areas:

- ☐ PWRP 2 - Plastic Waste Collection & Recycling
- ☐ TA 1.1 - Thermal energy generation
- ☒ TA 1.2 - Renewable Energy generation
- ☐ TA 10.1 - Fugitive emissions from oil and gas
- ☒ TA 13.1 - Solid waste and Wastewater
- ☐ TA 13.2 - Manure
- ☐ TA 14.1 - Afforestation and reforestation (except REDD & REDD+)

- ☐ TA 14.1 - Afforestation and reforestation (only REDD & REDD+)
- ☐ TA 15.1 - Agriculture (With Model Validation)
- ☐ TA 15.1 - Agriculture (Without Model Validation)
- ☐ TA 16.1 Carbon Capture and Storage ☐ TA 2.1 - Energy Distribution
- ☒ TA 3.1 - Demand Side Energy Efficiency
- ☐ TA 4.1 - Cement and Lime Production
- ☐ TA 4.n - Waste Heat Recovery & Fuel Switch ☐ TA 5.1 - Chemical Industry
- ☐ TA 5.2 - Caprolactam, Nitric and Adipic Acid ☐ TA 7.1 - Transport
- ☐ TA 8.1 - Mining/Mineral Production
- ☐ TA 9.1 - Aluminium and magnesium production
- ☐ TA 9.2 - Iron, Steel and Ferro-alloy Production
- ☐ TA PWRP 1 - Plastic Waste Collection & Recycling

Issue Date 29-12-2025

Expiry Date

29-12-2026

Vikash Kumar Singh
Director - Compliance

Carbon Check (India) Private Limited

Certificate of Competency

Seshan Ranganathan

has been qualified as per CCIPL's internal qualification procedures in accordance with the requirements of CDM AS, A 6.4 AS/ ISO/IEC14065:2020, ISO/IEC 17029:2019 and other applicable GHG programs:
for the following functions and requirements:

- | | | | |
|---|--|--|---|
| ☐ Management Function | ☐ Administrative | ☒ Validator/Verifier | ☒ Team Leader |
| ☐ Acting Team Leader | ☒ Technical Expert | ☒ Technical Reviewer | ☐ Trainee Assessor |
| ☐ Assessor | ☒ Regional Expert | ☐ Gender Expert | ☐ Financial Expert |
| | <div style="border: 1px solid black; padding: 2px; display: inline-block;">India</div> | | |
| ☐ Plastic Waste Expert | ☐ Health Expert | ☐ CCB Expert | ☐ SDG Expert |
| ☐ Expert Social Aspect | ☐ Expert Environmental Aspect | ☐ Internal Auditor | ☐ Legal Expert |
| ☐ FOEN Approved Technical Expert | ☐ FOEN Approved Quality Officer | | |

in the following Technical Areas:

- ☐ PWRP 2 - Plastic Waste Collection & Recycling
- ☒ TA 1.1 - Thermal energy generation
- ☒ TA 1.2 - Renewable Energy generation
- ☐ TA 10.1 - Fugitive emissions from oil and gas
- ☒ TA 13.1 - Solid waste and Wastewater
- ☒ TA 13.2 - Manure
- ☐ TA 14.1 - Afforestation and reforestation (except REDD & REDD+)

- ☐ TA 14.1 - Afforestation and reforestation (only REDD & REDD+)
- ☐ TA 15.1 - Agriculture (With Model Validation)
- ☐ TA 15.1 - Agriculture (Without Model Validation)
- ☐ TA 16.1 Carbon Capture and Storage ☐ TA 2.1 - Energy Distribution
- ☒ TA 3.1 - Demand Side Energy Efficiency
- ☐ TA 4.1 - Cement and Lime Production
- ☐ TA 4.n - Waste Heat Recovery & Fuel Switch ☒ TA 5.1 - Chemical Industry
- ☐ TA 5.2 - Caprolactam, Nitric and Adipic Acid ☐ TA 7.1 - Transport
- ☐ TA 8.1 - Mining/Mineral Production
- ☐ TA 9.1 - Aluminium and magnesium production
- ☐ TA 9.2 - Iron, Steel and Ferro-alloy Production
- ☐ TA PWRP 1 - Plastic Waste Collection & Recycling

Issue Date 12-12-2025

Expiry Date

12-12-2026

Vikash Kumar Singh
Director - Compliance

APPENDIX 5: FINDINGS LOG

Table 1. Remaining FAR from validation and/or previous verification

FAR ID	01	Section no.		Date: 10/07/2025
Description of FAR				
“Ex-ante Emission Reductions provided on the Verra page and in the PDMR are inconsistent. The estimated ERs are to be made consistent between the project documents and project activity web page. VVB to verify the same in the subsequent verification” With regard to the above FAR raised during joint validation and verification, the project participant is requested to ensure that the updated version of uploaded to the Verra registry's project page.				
Project proponent response				Date: 01/08/2025
The current ex-ante emission reductions number on the Verra page of 9,171,123 tCO₂e matches the latest PDMR submitted to Verra dated 1st August 2024. Upon submission of an updated PD to Verra, we will request the ex-ante emissions to be updated.				
Documentation provided by project proponent				
VVB assessment				Date: 07/08/2025
The VVB has reviewed the ex-ante emission reduction figure listed on the Verra Registry project page (9,171,123 tCO₂e over the first crediting period) and confirms that it matches the value stated in the most recent PDMR dated 16/07/2024. The PP has updated the PD due to the project's transition from the previously applied methodology VMR0006 v1.2 to the updated VM0050 v1.0 and will request Verra to update the estimated ER on the project page upon submission of updated PD, and VVB to verify the same in the subsequent verification. FAR 01 from this verification is raised.				

FAR ID	02	Section no.		Date: 10/07/2025
Description of FAR				
“For the current project activity instance, the Alluri Seetharama Raju district has been covered in the baseline survey. A baseline survey has to be conducted before the implementation of any new PAIs in compliance with the applied methodology section 9.1.” With regard to the above FAR raised during joint validation and verification, the project participant is requested to ensure that a baseline survey is conducted before implementation of any new project activity instance and provide records of such survey performed and related supportive documents.				
Project proponent response				Date: 01/08/2025

In Section 3.4 of the Monitoring Report, it has been explicitly stated that baseline surveys for Instance 2 were completed prior to distribution between 12th December 2023 and 6th January 2024. The start date of distribution for the instance was 7th January 2024, which can be verified from the database “ICS-01 Distribution Database M2”.

After distribution began in Instance 2, Verra released VM0050 v1.0 in October 2024. To transition from the earlier methodology (VMR0006 v1.2) to VM0050 v1.0, an additional 154 baseline surveys were conducted between 22nd November and 16th December 2024, and again in June 2025. These surveys were necessary to comply with VM0050's additional requirements, which included two new questions (and Kitchen Performance Tests) which were not covered in the initial baseline surveys conducted.

Documentation provided by project proponent

VVB assessment

Date: 07/08/2025

The VVB reviewed the baseline survey documentation and confirms that the initial surveys were conducted between 12 December 2023 and 6 January 2024, prior to the start of distribution on 7 January 2024, in line with VMR0006 v1.2. Following the adoption of VM0050 v1.0, the PP conducted additional baseline surveys in November-December 2024 and June 2025 to address the updated methodological requirements, including two new survey questions and KPTs.

The VVB concludes that the baseline surveys were implemented appropriately and in compliance with Section 9.1 of VM0050, and that the FAR has been satisfactorily addressed with respect to the current verification. For subsequent verifications, VVV to verify that a baseline survey has been conducted before the implementation of any new PAIs. FAR 02 from this verification is raised.

Table 2. CLs from this verification

CL ID	01	Section no.	1.12 MR	Date: 10/07/2025
Description of CL				
In section 1.12 of the MR, 1. The Monitoring Report mislabels the SDG table, using a title inconsistent with the required “Sustainable Development Contributions” per the VCS MR template. The Project Proponent shall clarify the reason for this deviation and revise the title. 2. On SDG 1.2.1 (Proportion of population living below the national poverty line, by sex and age), while time savings and shifts in daily activities are reported, direct evidence of income increase or measurable poverty reduction is not provided. The Project Proponent (PP) claims that time saved from firewood collection supports income-generating work, but no data on actual income changes (e.g., before vs. after ICS adoption) is presented. PP shall clarify the causal link between ICS use and poverty reduction with supporting economic data where possible. Additionally, SDG Indicator 1.2.1 requires disaggregated reporting. However, the response lacks sex- and age-specific data, which limits alignment with the SDG indicator metadata. PP shall clarify how such disaggregation is being addressed in the monitoring approach. 3. On SDG 1.4.1 (Proportion of the population living in households with access to basic services); the response reports the number of ICS distributed but does not quantify the proportion of the population that gained access, as required by the indicator. It lacks baseline access data, population context, and disaggregation by sex, age, or location. While cleaner cooking qualifies as a basic service, the contribution needs to be framed against the indicator definition. The Project Proponent (PP) shall clarify the proportion reached, provide disaggregated data, and reference supporting evidence. 4. The project reports increased time availability for education (45.3% in MP2, 53.9% overall), but this does not demonstrate actual participation in formal or non-formal education or training, as required by SDG Indicator 4.3.1. Additionally, no sex or age disaggregation is provided, which is a metadata requirement. Stove-use guidance does not qualify as structured non-formal education. PP shall either provide data evidencing formal/non-formal participation, disaggregated by sex and age or Reframe this as a project-specific indicator under SDG 4.3. 5. SDG 4.4.1: The reported activity does not align with the intent of the indicator, which measures the proportion of youth and adults with ICT skills by type. The contribution is limited to employee training without providing disaggregated data by age, sex, or ICT skill type. PP shall either revise the contribution to align with the indicator requirements or report this under a project-specific indicator. 6. SDG 5.4.1: Proportion of time spent on unpaid domestic and care work, by sex, age and location; the reported contribution does not align with the official indicator, which requires quantified time-use data specific to unpaid domestic and caregiving activities, disaggregated by sex, age, and location. The data provided reflect general time reallocation but not time spent on the relevant categories. PP shall clarify the methodology used to assess unpaid domestic and care work and consider reporting this under a project-specific indicator if alignment with SDG 5.4.1 cannot be demonstrated. 7. The contribution under SDG 8.5.2 is not fully aligned with the indicator, as it lacks disaggregation by sex, age, and disability status. This breakdown is essential for compliance with the SDG framework. The PP should either provide disaggregated data or reclassify the contribution as a project-specific indicator.				
Project proponent response				Date: 01/08/2025

1. The SDG table has been renamed in line with the VCS Monitoring Report Template v4.4
2. The indicator 1.2.1 has been removed
3. Additional data has been added into SDG 1.4 including the type and percentage of users using traditional stoves in the pre-project scenario. The indicator 1.4.1 has also been replaced as described below.

To address Points 3-7 of CL01, the indicators 1.4.1, 4.3.1, 4.4.1, 5.4.1 and 8.5.2 have been replaced with project specific indicators. These indicators directly align with the monitoring approach and outcomes described in section 1.12 of the Monitoring Report. Also see CAR ID02 for other changes to section 1.12 of the Monitoring Report.

Documentation provided by project proponent

VCS4457 Monitoring Report MP2 v1.1

VVB assessment

Date: 07/08/2025

For all SDGs: Under 'current project contributions' and 'contributions over project lifetime' the reported indicator values are not presented in similar way. PP is requested to modify accordingly. Also, PP is requested to include achieved contribution for the current MP and over the project lifetime for all SDGs other than SDG 13 in the ER calculation spreadsheet.

PP has updated and defined 'Project Specific Indicator: Proportion of beneficiaries able to spend more time contributing to the main family income, spending time with family, and helping others in the community'. This aligns with SDG 5.4 which focuses on recognizing and valuing unpaid care and domestic work and acceptable.

Hence this CL is closed.

CL ID	02	Section no.	2.1.4 MR	Date:	10/07/2025
Description of CL					
There is an inconsistency between Section 3.1, which reports that four stoves were damaged and repaired, and Section 2.1.4, which states that no grievances were raised during the monitoring period. The Project Proponent (PP) should clarify why these damages did not result in any recorded grievances. Specifically, the PP should explain whether the affected users were aware of the grievance mechanism, whether they raised informal or verbal concerns that were addressed without formal documentation, and why such instances were not captured in the grievance log. This clarification is necessary to assess the effectiveness and accessibility of the project's grievance redress mechanism.					
Project proponent response				Date:	01/08/2025

The damages were raised and recorded in the grievance log, which has been checked and verified by the VVB during the project site visit. Section 2.1.4 has been updated to include details of these repairs including the ICS QR code, date of repair, parts replaced and maintenance team member responsible for conducting the service. No formal complaints were made by the users of these stoves and therefore we did not originally record this in the Monitoring Report (and only provided it in the “ICS-01 Monitoring Database M2”). Going forward all repairs and replacements will form part of the grievance log.

Families are made aware of the grievance mechanism during initial community stakeholder meetings and at the point of ICS distribution. Contact details of the local area coordinator are provided to beneficiaries which ensures consistent support and communication when needed.

Documentation provided by project proponent

- Grievance Log Folder
- VCS4457 Monitoring Report MP2 v1.1

VVB assessment

Date: 07/08/2025

VVB reviewed the updated section 2.1.4 and confirmed it is consistent with section 3.1 of MR. Further, VVB has reviewed grievance log and confirmed that the stove damages were indeed recorded. Therefore CL 02 is closed.

CL ID	03	Section no.	3.4 MR	Date:	10/07/2025
Description of CL					
In section 3.4 of the MR, 1. The Monitoring Report lacks two key elements required by the MR template: it does not provide the percentage change in baseline emissions between Instance 1 and Instance 2, and it does not identify the updated sections of the Project Description (PD) or confirm if none were required. The Project Proponent (PP) shall clarify by including the percentage change in baseline emissions with supporting data, and identifying the PD sections updated due to the reassessment, or provide justification if no updates were necessary. 2. The baseline surveys (1472 baseline surveys) for the second project instance lack essential details on sampling methodology, population size, and achieved confidence and precision levels, as required by CDM AMS-II.G and VM0050 (minimum 90% confidence, 10% precision). PP shall provide this information to demonstrate methodological compliance. 3. for the second project instance 154 Kitchen Performance Tests (KPTs) were performed and lack essential details on sampling methodology, population size, and achieved confidence and precision levels, as required by CDM AMS-II.G and VM0050 (minimum 90% confidence, 10% precision). PP shall provide this information to demonstrate methodological compliance. 4. In Section 3.4, Baseline Reassessment, paragraph 5, the report states that a baseline survey was conducted for Instance 2 to measure average baseline fuel consumption. However, as per the VM0050 v1.0 Corrections & Clarifications document (February 3, 2025), project proponents (PP) are required to cross-check baseline fuel consumption results using reputable external sources such as government publications, peer-reviewed studies, or official data. This requirement has not been fulfilled in the current submission. The PP shall address this gap and provide the necessary cross-verification in line with the guidance.					
Project proponent response				Date:	
				01/08/2025	

1. The percentage change in baseline emissions between Instance 1 and 2 has been added into paragraph 7 of section 3.4. As the baseline for Instance 1 remains unchanged and still relevant, no changes to the PD have been made as a result of the baseline reassessment. This has been justified in the first paragraph of section 3.4
2. In paragraph 2 of section 3.4, the use of random sampling has been described. Reference to the sampling section of the report (section 4.3.1) has been added to provide additional clarity. Within section 4.3.1, QA/QC checks for the representative baseline parameter Hh_i have been included to demonstrate that the baseline survey meets the 90/10 confidence/precision requirement.
3. In section 4.3.1, details of the random sampling procedure used to select the Instance 2 KPT households has been described. Statistical analysis of the parameters $BC_{b,i,y}$ and $BC_{p,j,k,y}$ for Instance 2 has also been presented in this section to demonstrate that the KPTs conducted meet the 90/10 confidence/precision requirements of the VM0050 and CDM methodology.
4. The baseline fuel consumption for Instance 2 has been cross-checked with an external study conducted in the project region by Koshish Sustainable Solutions, an independent company specialising in biomass cookstove testing. Details of this study, including comparisons to the Instance 2 KPT results have been presented and explained in section 3.4. The project report submitted by Koshish Sustainable Solutions has been provided separately to the VVB.
Note that there were no official government reports or publications published after 2013 to use for comparison. This is discussed further in section 3.4 of the Monitoring Report.

Documentation provided by project proponent

- Baseline Survey Instance 2 Random Sample
- Baseline Sample Size + Precision
- KPT Instance 2 Random Sample
- Koshish KPT Report 13-02-25
- VCS4457 Monitoring Report MP2 v1.2

VVB assessment

Date: 07/08/2025

1. The VVB has reviewed the updated paragraph 7 of section 3.4 and confirmed percentage change in baseline emissions between Instance 1 and 2 has been added. Further, VVB reviewed the justification in the first paragraph of section 3.4 and confirmed that not revising the PD is reasonable.
2. The VVB has reviewed the updated section of MR and confirmed sampling methodology, population size, and achieved confidence and precision levels are described in MR.
3. PP has updated section 4.3.1 of the revised MR and adequately demonstrated demonstrate that the KPTs conducted meet the 90/10 confidence/precision requirements of the VM0050 and CDM methodology
4. VVB reviewed the Koshish KPT Report for Instance 2 baseline fuel consumption. VVB considers this, in the absence of any credible government data that are not outdated, sufficient to comply with methodological requirement.

Hence this CL is closed.

CL ID	04	Section no.	4.3.1 MR	Date: 10/07/2025
Description of CL				
1. In Section 4.3.1, Sample Design/Sizing and Field Data, while the MR states that KPTs were conducted, it lacks details on how households were selected and does not confirm whether the sampling followed the simple random sampling (SRS) method as required by the PD. This omission raises concerns about the representativeness and validity of the KPT data, which is critical for estimating fuel savings and emission reductions. The PP shall include a clear explanation of the sampling process used, specifying the technique applied (e.g., SRS), the population from which the sample was drawn, and how randomization was implemented in practice. 2. In Section 4.3.1, Sample Design/Sizing and Field Data (fourth paragraph), the report presents results from 6 Water Boiling Tests (WBTs) but does not include a statistical summary to demonstrate compliance with the methodology's 90% confidence and 10% relative precision requirement. The Project Proponent shall provide the mean, standard deviation, and calculated confidence interval or relative precision to confirm that the reported thermal efficiency meets the required statistical standard. 3. In Section 4.3.1.1, Sample Size, the report cites a population size of 115,601 for the monitoring period and uses this to calculate the required sample size for $n_{j,k,y}$ based on a 90/10 confidence/precision level. However, it does not include a reliability check or analysis for the proportion of commissioned project devices of type j in batch k operating during year y. The Project Proponent shall clarify and confirm how this proportion was assessed, and whether reliability thresholds were met as per methodological requirements.				
Project proponent response				Date: 01/08/2025
1. To select the households to participate in the KPT for Instance 2, a simple random sampling approach was used. This has been presented in section 4.3.1 of the Monitoring Report. Specifically, each household from the population was assigned a unique identifier. Using Excel's RANDARRAY function, a sample of 95 random unique identifiers were generated, representing a sample of the population for which KPTs were conducted. This approach guaranteed that each household had an equal chance of being selected, resulting in a representative and unbiased sample. 2. Statistical analysis of the WBTs has been added into section 4.3.1 of the Monitoring Report. The mean, standard deviation, confidence interval and precision have been calculated and included in the section. The precision of the 18 samples is calculated as 0.27% and therefore the WBT results meet the 90/10 confidence/precision requirements of the methodology. A detailed breakdown of the calculation is described in section 4.3.1. 3. A reliability check has been added into section 4.3.2, following the calculation of the minimum sample size for the parameter $n_{j,k,y}$. The mean and standard deviation of $n_{j,k,y}$ has been calculated as 1.0 and 0 respectively, as all ICS samples monitored were found operational owing to regular maintenance and repairs being conducted. As the standard deviation is 0, the monitoring survey meets the 90/10 confidence/precision threshold of the methodology.				
Documentation provided by project proponent				
• KPT Instance 2 Random Sample • Monitoring Period 2 Random Sample • MP2 Sample Size + Precision • VCS4457 Monitoring Report MP2 v1.2				
VVB assessment				Date: 07/08/2025

1. VVB reviewed the updated section 4.3.1 of the Monitoring Report and confirms the sampling followed the simple random sampling (SRS) method as required by the PD. CL 04-01 is closed.
2. VVB reviewed the updated details for the WBTs, including mean, standard deviation, confidence interval, and calculated precision (0.27%), in Section 4.3.1 and confirmed it is described appropriately. Therefore CL 04 – 02 is closed.
3. The VVB reviewed updated section of the MR and confirmed that the reliability check has been included, and the reliability threshold have been met as per methodological requirements. Therefore CL 04 – 03 is closed.

CL ID	05	Section no.	MR 4.2	Date: 10/07/2025
Description of CL				
The measurement method, to determine the value for parameter $\eta_{new,avg,y}$, is as per CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities v9.0 (option 1 of the applied methodology VM0050, v1.0); however it is not clear how the required sample size has been attained and whether this complies with para 14 of the aforementioned sampling standard which states, "If the parameter of interest is a numeric mean value (i.e. not a proportion or percentage) the Student's t-distribution shall be used if the resulting sample size is less than 30". PP is requested to clarify on this and provide reliability test (to demonstrate 90/10 confidence and precision levels are met) and sample size calculation sheet.				
Project proponent response				Date: 01/08/2025
To measure the decrease in thermal efficiency of the project device (for y=1 onwards) either option a) Standard Water Boiling Test campaigns or b) a linear decrease approach can be used in line with the VM0050 v1.0 methodology. For y=1, we selected option a), the results of which are presented in Appendix 2 of the Monitoring Report. Using this option, there is no necessity to align with the CDM Standards, but the test must instead be conducted "following national standards (where available) or international standards or guidelines". The sample size has therefore been determined in line with the Water Boiling Test v4.2.3 and ISO standards, which do not explicitly state a minimum sample size, although 3 stoves (9 tests) is the industry standard. Instead, Appendix 5 of the Water Boiling Test v4.2.3 discusses different statistical approaches to ensure reliability in the data measured. To ensure reliability in the measurements, we have opted to use the CDM Standard for Sampling and Surveys. The Student t-distribution has been adopted as the sample size is less than 30. The mean and standard deviation have been calculated from the test results and then used to calculate the precision, assuming 90% confidence. From the 18 samples, the precision was calculated as 0.27%, therefore demonstrating that a 90/10 confidence/precision level has been met. A detailed breakdown of this analysis is presented in section 4.3.1 of the Monitoring Report.				
Documentation provided by project proponent				
VCS4457 Monitoring Report MP2 v1.2				
VVB assessment				Date: 07/08/2025

The Project Proponent has clarified that the parameter $\eta_{new,avg,y}$ was determined using Option (a) of VM0050 v1.0 through the Standard Water Boiling Test (WBT), in line with international testing standards. While the CDM sampling standard is not mandatory under this option, the PP applied the Student's t-distribution given the sample size was below 30, as per paragraph 14 of the CDM Standard for Sampling and Surveys. Based on 18 test results, the precision was calculated at 0.27% with 90% confidence, demonstrating that the 90/10 confidence and precision levels are met. The analysis is presented in Section 4.3.1 of the Monitoring Report. The approach and supporting evidence are acceptable. Therefore CL 05 is closed.

Table 3. CARs from this verification

CAR ID	01	Section no.	1.10 and 1.11 of MR	Date: 10/07/2025
Description of CAR				
In Sections 1.10 and 1.11 of the report, the response provided is "No"; therefore, the follow-up clause "If yes..." is not applicable and shall be removed to maintain clarity and relevance. The PP is requested to rectify as required and remove any instructional text of from the template throughout the document.				
Project proponent response				Date: 01/08/2025
This has been corrected throughout the document and removed accordingly.				
Documentation provided by project proponent				
VCS4457 Monitoring Report MP2 v1.2				
VVB assessment				Date: 07/08/2025
The VVB has reviewed the MR and confirmed that all instructional text has been removed throughout the document. Therefore, CAR 01 is closed.				

CAR ID	02	Section no.	1.12 MR	Date: 10/07/2025
Description of CAR				
In section 1.12 of the MR, 1. Section 1.12 does not meet VCS MR template requirements. It lacks a brief summary of SD activities conducted during the monitoring period, does not specify activity locations, and fails to explain how these activities led to the SDG contributions in Table 1. The PP shall revise to include these elements as required. 2. The reported data under SDG Indicator 7.1.2 lacks the required calculation of the proportion of the population using clean fuels and technology. While absolute figures are provided, the monitoring report must also present this as a percentage of the total population in the intervention area. PP shall update the report accordingly to align with SDG 7.1.2 requirements. 3. The indicator 7.a.1 is defined as International financial flows covered through Organisation for Economic Co-operation and Development (OECD) and International Renewable Energy Agency (IRENA) as per United Nations Department of Economic and Social Affairs Statistics. PP is to justify using carbon credit market participation by this project as the contribution linked with this indicator. Also PP has neither reported any measured contribution for the current monitoring period nor for over project lifetime, which is to be clarified. 4. The report presents aggregate firewood savings but does not align with the full scope of SDG Indicator 12.2.2, which requires material consumption to be expressed per capita and per unit of GDP. The Project Proponent should revise the documentation to include firewood savings data normalized per capita and, where possible, per GDP unit. 5. The reported forest area preserved due to reduced firewood use is a project-specific proxy, not the official SDG 15.1.1 indicator. PP shall clearly state this, justify the methodology and assumptions used to convert firewood savings into hectares, and explain how these estimates reflect actual conservation outcomes. Additionally, SDG Targets 15.2 and 15.3 are referenced without defined indicators or supporting data. PP shall revise the monitoring documentation to align with official SDG indicators.				
Project proponent response				Date: 01/08/2025

1. Section 1.12 has been revised to include a summary of the activities during the monitoring period, including locations and how the activities lead to the SDG contributions presented in Table 3.
2. In addition to the absolute number, the project beneficiaries have been presented as a proportion of the Alluri Sitharama Raju population to align with the requirements of SDG 7.1.2
3. In line with the definition of Indicator 7.a.1 (according to https://worldbank.github.io/sdg-metadata/metadata/en/7-a-1/#SDG_INDICATOR), the indicator has been removed as it does not include private financial flows. The impact on SDG 7 is now measured only with indicator 7.1.2 as presented in Table 3 of the Monitoring Report.
4. The indicator 12.2.2 has been replaced with a project-specific indicator. The contributions for the monitoring period directly align with the updated indicator and therefore have not been revised or normalised per capita.
5. The indicators under SDG 15 have been replaced with a single project-specific indicator which aligns with the outcomes described for the monitoring period.
To define the contribution to SDG 15, calculated firewood savings have been converted into hectares as presented in spreadsheet "ICS-01 Fuel Savings M2". Specifically, the mass of wood has been first converted into volume using the height, radius and density of a locally widespread tree species, silver oak. Note parameter values in the spreadsheet are from published literature and crosschecked by the local team experienced within the project region. Once a volume saved has been determined, the area is calculated using the average estimated tree spacings across the project area. To ensure a conservative estimate:
 - A 50% adjustment is applied to the final area saved to account for wood which is not cut down e.g. downed deadwood collected from the forest floor
 - The pre-project fuel consumption has been capped in line with VM0050 (11.53kg is used for Instance 1 versus 13.22kg measured in our KPTs)
 - The efficiency of the traditional stove has been taken as 15%

Documentation provided by project proponent

- ICS-01 Fuel Savings M2
- VCS4457 Monitoring Report MP2 v1.2

VVB assessment

Date: 07/08/2025

1. The revised Section 1.12 now includes a summary of sustainable development activities conducted during the monitoring period, specifying activity locations and clearly explaining the linkages between activities and the SDG contributions presented in Table 3. This adequately addresses the VCS MR template requirements. CAR 02-01 is closed.
2. The Monitoring Report has been updated to present the number of beneficiaries as a proportion of the Alluri Sitharama Raju district population, in addition to absolute figures. This aligns with the SDG 7.1.2 reporting requirements. CAR 02-02 is closed.
3. The Project Proponent has removed Indicator 7.a.1, as private financial flows such as carbon revenue are not covered under the official SDG definition (OECD/IRENA sources). The contribution to SDG 7 is now represented solely through Indicator 7.1.2, which is appropriate. CAR 02-03 is closed.
4. The Project Proponent has replaced SDG 12.2.2 with a project-specific indicator reflecting firewood savings achieved during the monitoring period. While not expressed per capita or GDP unit, the approach is justified since the project indicator more directly reflects measurable project outcomes. The revision is acceptable. CAR 02-04 is closed.
5. The use of a single project-specific indicator for SDG 15 is justified. The methodology for converting firewood savings to forest area preserved has been described in detail and is supported by conservative assumptions, literature-based parameters, and adjustments to ensure conservativeness (e.g., 50% reduction for deadwood, capped baseline consumption, and efficiency factors). The applied approach is transparent and reasonable. CAR 02-05 is closed.

CAR ID	03	Section no.	2.2 MR	Date: 10/07/2025
Description of CAR				
Section 2.2 of the MR does not align with the VCS Monitoring Report template, which requires a table listing validation-identified risks and mitigation measures applied during the monitoring period. As a new instance has been added, the risk profile may differ from that in the PD. Where no risks are identified, a justification is required. PP shall revise section 2.2 accordingly to ensure full compliance with the template.				
Project proponent response				Date: 01/08/2025
The risk profile of households in Instance 2 has not changed from that in Instance 1 due to the consistency in stove technology, project approach and community profile. Section 2.2 has been updated to include justification for this, in compliance with the VCS Monitoring Report Template v4.4.				
Documentation provided by project proponent				
VCS4457 Monitoring Report MP2 v1.2				
VVB assessment				Date: 07/08/2025

VVB has reviewed the updated section 2.2 of the revised MR and confirmed it now provides adequate justification of applying risk assessment described in relevant section of the PD. Therefore CAR 03 is closed.

CAR ID	04	Section no.	3.2.2 MR	Date: 10/07/2025
Description of CAR				
Section 3.2.2 does not align with the VCS Methodology Change and Requantification Procedure v4.0, which states that project proponents applying this procedure are not required to submit a project description deviation at subsequent verifications. PP shall revise this section to reflect this requirement.				
Project proponent response				Date: 01/08/2025
The text in section 3.2.2 has been revised to state that no project description deviation has been applied. This now aligns with the VCS Methodology Change and Requantification Procedure v4.0				
Documentation provided by project proponent				
VCS4457 Monitoring Report MP2 v1.2				
VVB assessment				Date: 07/08/2025
VVB has reviewed the updated section and confirmed that it aligns with the VCS methodology change and Requantification procedure v4.0. Therefore CAR 04 is closed.				

CAR ID	05	Section no.	4.2 MR	Date: 10/07/2025
Description of CAR				
In section 4.2 of the MR,				
Data / Parameter: $N_{j,k,y}$ – Number of Commissioned Project Devices of Type j from Batch k in Year y. The Value Monitored field does not specify the exact number of improved cookstoves (ICS) distributed during the monitoring period. PP shall provide the total number of distributed devices (summed value). Data / Parameter: $n_{j,k,y}$ – Proportion of Commissioned Project Devices of Type j from Batch k Still in Use in Year y. The Value Monitored field does not report the actual monitored value or an indicative range for $n_{j,k,y}$ as derived from the monitoring survey. PP shall report the calculated or average value observed during the monitoring period, along with basic descriptive statistics such as the mean and standard deviation. Data / Parameter: $f_{NRB,y}$ – Fraction of Non-Renewable Biomass: The report does not include QA/QC procedures for the parameter $f_{NRB,y}$ (Fraction of Non-Renewable Biomass), which is critical to emission reduction estimates. The Project Proponent shall: (i) disclose data sources and estimation methods for NRB and RB, including geographic scope and stakeholder input; and (ii) provide QA/QC procedures for verifying biomass supply data, validating assumptions, and cross-checking input values against authoritative sources.				
project proponent response				Date: 01/08/2025

1. The total number of stoves distributed over the monitoring period, 115,601, has been added into the “Value Monitored” field for $N_{j,k,y}$. 2. The mean value monitored for the period (1.0) and the standard deviation (0) have been added in the “Value Monitored” field for the parameter $n_{j,k,y}$. 3. For the parameter $f_{NRB,y}$: i) The “Monitoring Equipment” field has been updated to list the data sources and estimation methods used for NRB and RB. These include KPTs conducted in Instance 2 to measure household fuel consumption as well as use of published literature and government reports. Note the comprehensive list of data sources is provided in the spreadsheet “ICS-01 fNRB Calculation Instance 2” ii) The field “QA/QC procedures to be applied” has been updated to demonstrate what crosschecks and data sources have been used to ensure the reliability of the NRB and RB inputs. These have also been presented in the respective spreadsheets: “ICS-01 fNRB Calculation Instance 1” and “ICS-01 fNRB Calculation Instance 2”.	
Documentation provided by project proponent	
VCS4457 Monitoring Report MP2 v1.2	
VVB assessment	Date: 07/08/2025
1. VVB reviewed the updated section of MR and confirmed the total number of stoves distributed over the monitoring period, has been added into the “Value Monitored” field for $N_{j,k,y}$. 2. VVB reviewed the updated section of MR and confirmed the actual monitored value for parameter $n_{j,k,y}$ has been added. 3. VVB reviewed the updated section and confirmed QA/QC procedures for the parameter $f_{NRB,y}$, data sources and estimation methods for NRB and RB, including geographic scope and stakeholder input. Hence, CAR 05 is closed.	

CAR ID	06	Section no.	4.3.1 MR	Date: 10/07/2025
Description of CAR				
In section 4.3.1 of the MR, 1. Under ‘Sample Design/Sizing and Field Data’ (second paragraph), the report does not present the baseline fuel consumption values (BCb,i,y) and the project fuel consumption values (BCp,j,y), nor does it include baseline KPT results, sampling information, or testing design details. The PP shall revise this section to clearly provide the BCb values used, along with corresponding KPT data and methodological details for each instance. 2. Under ‘Sample Size’, the report cites a population size of 115,601 and calculates the required sample size for $n_{j,k,y}$ using a 90/10 confidence/precision level. However, it does not present a reliability check or analysis for the proportion of commissioned project devices of type j in batch k operating during year y. The Project Proponent shall clarify and confirm whether the required reliability thresholds were met in accordance with the methodological requirements.				
Project proponent response				Date: 01/08/2025

1. Section 4.3.1 has been updated to clearly state the values of $BC_{b,i,y}$ and $BC_{p,j,k,y}$ used for both project Instances, which correspond to the KPT results. The section also includes details of the simple random sampling procedure used to select the KPT households and details on how the KPT was conducted including exact procedure and timeframes. Statistical analysis of the parameters $BC_{b,i,y}$ and $BC_{p,j,k,y}$ for Instance 2 has been presented in the section to demonstrate that the KPTs conducted meet the 90/10 confidence/precision requirements of the VM0050 and CDM methodology.
2. A reliability check has been added into section 4.3.2, following the calculation of the minimum sample size for the parameter $n_{j,k,y}$. The mean and standard deviation of $n_{j,k,y}$ has been calculated as 1.0 and 0 respectively, as all ICS samples monitored were found operational owing to regular maintenance and repairs being conducted. As the standard deviation is 0, the monitoring survey meets the 90/10 confidence/precision threshold of the methodology.

Documentation provided by project proponent

- Monitoring Period 2 Random Sample
- MP2 Sample Size + Precision
- VCS4457 Monitoring Report MP2 v1.2

VVB assessment

Date: 07/08/2025

1. VVB reviewed the updated section and confirmed the baseline fuel consumption values ($BC_{b,i,y}$), project fuel consumption values ($BC_{p,j,y}$), baseline KPT results, sampling information and testing design details are described.
2. VVB reviewed the updated MR and confirmed that a reliability check has been added. Further, VVB confirmed the required reliability thresholds were met in accordance with the methodological requirements.

Hence, CAR 06 is closed.

CAR ID	07	Section no.	ER sheet	Date:	10/07/2025
Description of CAR					
The baseline, project, leakage, and emission reduction values for both monitoring periods—separated by vintage—are provided in the emission reductions calculation sheet. However, the values are inconsistent within the different tabs of the spreadsheet “ICS-01 Emission Reductions M2” (e.g., Reduced ER per vintage). The Project Proponent shall correct these inconsistencies to ensure internal consistency across all tabs and alignment with the Monitoring Report.					
Project proponent response				Date:	01/08/2025

The values reported have been double checked and line up with the values in the file “ICS-01 Emission Reductions M2”.

Note the reported emissions reductions in Table 2 of the Monitoring Report, split by monitoring period, come directly from the “ICS-01 Emission Reductions M2” file, specifically the third table in the sheet named “Emission Reductions Summary”.

The emission reductions in section 5.4 of the Monitoring Report, which are split specifically by vintage, come from instead the fourth table in the same sheet above.

The reported emission reductions in the subsequent sheets of the “ICS-01 Emission Reductions M2” file, specifically calculate the emission reductions split by monitoring period and vintage period. Note the values calculated in sheets “EC Calc”, “Instance 1 MP2”, “Instance 2 MP2”, “Instance 1 – 2023 split” and “Instance 1 – 2024 split” represent inputs which feed directly into the “Emission Reductions Summary” sheet.

Documentation provided by project proponent

VCS4457 Monitoring Report MP2 v1.2

VVB assessment

Date: 07/08/2025

VVB reviewed the emission reduction values for both monitoring period separated by vintage in ER sheet’s all the tabs and confirmed it is consistent. Therefore CAR 07 is closed.

CAR ID	08	Section no.	MR	Date: 10/07/2025
Description of CAR				
There are a few areas that require formatting corrections to align with the VCS MR template, particularly the date format along with editorial consistency and spacing issues throughout the report.				
Project proponent response				Date: 01/08/2025
Date formatting and editorial inconsistencies have been corrected throughout the document in line with the VCS Monitoring Report Template v4.4.				
Documentation provided by project proponent				
VCS4457 Monitoring Report MP2 v1.2				
VVB assessment				Date: 07/08/2025
VVB reviewed the updated document and confirmed Date formatting, editorial consistency and spacing is consistent throughout the document. Therefore CAR 08 is closed.				

CAR ID	09	Section no.	MR	Date:	10/07/2025
Description of CAR					
The ICS samples selected to conduct KPT as part of the monitoring activity for the current monitoring period were distributed outside the monitoring period. PP needs to justify how these samples represent the ICS population accounted for this monitoring period.					
Project proponent response				Date:	
				27/09/2025	

The monitoring period occurred prior to the release of VM00050 in October 2024. At that time, the project was registered under VMR0006 v1.2, which did not require Kitchen Performance Tests (KPTs). Therefore, no KPTs were conducted during the actual monitoring period.

Following the release of VM00050, we conducted baseline and project KPTs using paired samples. The approach of using the same household for the baseline and project KPTs, was to provide the most accurate method for measuring the change in fuel consumption. To ensure the results were also representative of the ICS population over the monitoring period, we conducted additional project KPTs in September 2025 using stoves from within the specific monitoring period.

Note that at the time of the latest (September 2025) KPTs, the average age of the stoves was 1.8 years, whereas the average age of the stoves across the full monitoring period was 0.8 years. This means the KPT results are conservative, as older project stoves typically demonstrate reduced performance (and greater emissions).

In conclusion, we have combined the project KPT results from both the paired samples as well as those from within the monitoring period, to output KPT data which is most representative of the ICS population for the specific monitoring period.

Documentation provided by project proponent

- ICS-01 KPT Instance 1
- ICS-01 KPT Instance 2

VVB assessment

Date: 28/09/2025

Additional project KPTs were conducted in September 2025 using stoves from within the monitoring period. At the time of the September 2025 KPTs, the average stove age was 1.8 years, compared to 0.8 years across the monitoring period. This makes the KPT results conservative.

Hence CAR 09 is closed.

Table 4. FARs from this verification

FAR ID	01	Section no.		Date: 07/08/2025
Description of FAR				
The estimated emission reductions are to be made consistent between the Verra project page and the project documents. VVB to verify the same in the subsequent verification.				
Project proponent response				Date: DD/MM/YYYY
Documentation provided by project proponent				
VVB assessment				Date: DD/MM/YYYY

FAR ID	02	Section no.		Date: 07/08/2025
Description of FAR				
PP to comply with para 6.2 of VMR0006 v1.2 in conjunction with correction 1 of Corrections & Clarifications to VMR0006 v1.0 and must conduct a simple random sampling where a survey covers two or more project activity instances and the population is homogeneous, and where the population is not homogenous, stratified random sampling must be used. Also, follow-up baseline surveys must be conducted at most every five years from the date of the last survey in control households that do not participate in the project. Moreover, baseline survey performed prior to crediting period renewal must be conducted as physical on-site visit. VVB to verify this during subsequent verifications. FAR 02 from this verification is raised.				
Project proponent response				Date: DD/MM/YYYY
Documentation provided by project proponent				
VVB assessment				Date: DD/MM/YYYY